

TECHNICAL REPORT



Process management for avionics – Electronic components capability in operation – Part 1: Temperature uprating



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TECHNICAL REPORT



Process management for avionics – Electronic components capability in operation – Part 1: Temperature uprating

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

ICS 03.100.50; 31.020; 49.060

ISBN 978-2-8322-5364-9

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COMPONENTS CAPABILITY IN OPERATION –****Part 1: Temperature uprating****FOREWORD**

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IEC TR 62240-1, which is a Technical Report, has been prepared by IEC technical committee 107: Process management for avionics.

This second edition cancels and replaces the first edition published in 2013. This edition constitutes a technical revision. This edition includes the following significant technical changes with respect to the previous edition:

- a) Revised the wording in 4.1 and the corresponding Figure 1 to reflect current industry practices.

The text of this Technical Report is based on the following documents:

CDTR	Report on voting
107/313/DTR	107/322/RVDTR

Full information on the voting for the approval of this Technical Report can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 62240 series, published under the general title *Process management for avionics – Electronic components capability in operation*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

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- replaced by a revised edition, or
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INTRODUCTION

Traditionally, industries that produced electronic equipment for ADHP (aerospace, defence and high performance) applications have relied on the military specification system for semiconductor device standards and upon manufacturers of military-specified devices as device sources. This assured the availability of semiconductor devices specified to operate over the temperature ranges required for electronic equipment in ADHP applications. In the past, several device manufacturers have exited the military market, resulting in the decreased availability of devices specified to operate over wide temperature ranges. Following are some typical ambient temperature ranges at which devices are marketed:

Military:	–55 °C to + 125 °C
Automotive:	–40 °C to + 125 °C
Industrial:	–40 °C to + 85 °C
Commercial:	0 °C to + 70 °C

If there are no reasonable or practical alternatives, then a potential response is for electronic equipment manufacturers to use devices at temperature ranges that are wider than those specified by the device manufacturer.

This document provides information on selecting semiconductor devices, assessing their capability to operate, and assuring their intended quality in the wider temperature ranges. It also reports the need for documentation of such usage.

This can be supported by exchanging technical information with the original device manufacturer.

Operation of the device beyond the manufacturer's limits can result normally in loss of warranty by the device manufacturer.

PROCESS MANAGEMENT FOR AVIONICS – ELECTRONIC COMPONENTS CAPABILITY IN OPERATION –

Part 1: Temperature uprating

1 Scope

This part of IEC 62240, which is a technical report, provides information when using semiconductor devices in wider temperature ranges than those specified by the device manufacturer. The uprating solutions described herein are considered exceptions, when no reasonable alternatives are available; otherwise devices are utilized within the manufacturers' specifications.

The terms “uprating” and “thermal uprating” are being used increasingly in avionics industry discussions and meetings, and clear definitions are included in Clause 3. They were coined as shorthand references to a special case of methods commonly used in selecting electronic components for circuit design.

This document describes the methods and processes for implementing this special case of thermal uprating. All of the elements of these methods and processes employ existing, commonly used best engineering practices. No new or unique engineering knowledge is needed to follow these processes, only a rigorous application of the overall approach.

Even though the device is used at wider temperatures, the wider temperatures usage will be limited to those that do not compromise applications performance and reliability, particularly for devices with narrow feature size geometries (for example, 90 nm and less). This document does not imply that applications use the device to function beyond the absolute maximum rating limits specified by the original device manufacturer and assumes that:

- device usage outside the original device manufacturers' specified temperature ranges is done only when no reasonable alternative approach is available and is performed with appropriate justification;
- if it is necessary to use devices outside the original device manufacturers' specified temperature ranges, it is done with documented and controlled processes that assure integrity of the electronic equipment.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC TS 62239-1, *Process management for avionics – Management plan – Part 1: Preparation and maintenance of an electronic components management plan*

3 Terms, definitions and abbreviated terms

3.1 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1.1

absolute maximum ratings

limiting values of operating and environmental conditions applicable to any semiconductor device of a specific type as defined by its published specification data, which should not be exceeded under the worst possible conditions

[SOURCE: IEC 60134:1961, Clause 4]

3.1.2

ambient temperature

temperature of the environment in which a semiconductor device is operating

3.1.3

case temperature

temperature of the surface of a semiconductor device package during operation

3.1.4

circuit element functional mode analysis

documented analysis that determines minimum ranges and maximums of all functional characteristics of the assembly with respect to the related functional parameters of devices being updated

3.1.5

COTS product

commercial off-the-shelf product

one or more components, assembled and developed for multiple commercial consumers, whose design and/or configuration is controlled by the manufacturer's specification or industry standard

Note 1 to entry: COTS products can include electronic components, subassemblies or assemblies, or top level assemblies. Electronic COTS subassemblies or assemblies include circuit card assemblies, power supplies, hard drives, and memory modules. Top-level COTS assemblies include a fully integrated rack of equipment such as raid arrays, file servers to individual switches, routers, personal computers, or similar equipment.

[SOURCE: IEC TS 62668-1:2016, 3.1.3]

3.1.6

device capability assessment

process of demonstrating that the device design is capable of providing the specified functionality and operation over the wider temperature range, for the required length of time

Note 1 to entry: It assumes that the device has been qualified to operate within its specified temperature range, and includes additional testing or analysis to evaluate expected performance at the wider temperature range. Device capability assessment includes both performance and application-specific reliability.

3.1.7

device quality assurance over the wider temperature range

additional testing or analysis required to assure that each individual device is capable of operating successfully in the required wider temperature range

3.1.8

device

component

material element or assembly of such elements intended to perform a required function

Note 1 to entry: A device may form part of a larger device.

[SOURCE: IEC 60050-151:2001, 151-11-20, modified – The term "component" has been added as a synonym to "device".]

3.1.8.1

semiconductor device

electrical or electronic device that is not subject to disassembly without destruction or impairment of design use

Note 1 to entry: It is sometimes called electronic part or electronic piece part or component or electronic component. Examples are diodes, integrated circuits, and transistors.

3.1.9

electronic equipment

any item, for example end item, sub-assembly, line-replaceable unit, shop-replaceable unit, or system produced by an electronic equipment manufacturer

3.1.10

environmental stress screening

ESS

highly accelerated stress screening

HASS

set of production process tasks consisting in applying to the equipment concerned, within the limits permitted by its design, particular environmental stresses in order – during manufacturing – to reveal and eliminate the largest possible number of extrinsic defects which, in all probability, would have appeared once utilisation had begun (early life failures)

[Source: IEC TS 62500:2008, 2.8, modified – The second term, HASS, has been added.]

3.1.11

highly accelerated test

HAT

highly accelerated life test

HALT

test during which the product or some of its component parts are subjected to environmental and/or operating stresses that are increased progressively to values far in excess of the specified values, up to the operating and/or destruction limits of the product

NOTE The rise in exposure time or number of cycles, whether or not associated with a combination of certain stresses raised to values close to or equal to the specification (or stresses whose nature is not specified) may meet the same targets as those of the highly accelerated tests, as defined in this document.

[Source: IEC TS 62500:2008, 2.10, modified – The second term, HALT, has been added and in the note "technical specification" has been replaced by "document".]

3.1.12

junction temperature

temperature of the active region of the device in which the major part of the heat is generated

[SOURCE: SEMATECH Dictionary of Semiconductor Terms:2012]

3.1.13

manufacturer-specified parameter limits

electrical parameter limits that are guaranteed by the device manufacturer when a device is used within the recommended operating conditions

3.1.14

manufacturer-specified temperature range

operating temperature range over which the component specifications, based on the component data sheet, are guaranteed by the electronic component manufacturer

Note 1 to entry: Manufacturer-specified temperature range is a subset of the recommended operating conditions.

3.1.15**parameter conformance assessment**

process for thermal uprating in which devices are tested to assess their conformance to the manufacturer-specified parameter limits over the target wider temperature range

3.1.16**parameter temperature characterisation**

process of determining the specification values of electrical parameters by testing samples over the manufacturer's specified temperature range

3.1.17**parameter temperature re-characterisation**

process for thermal uprating in which the device parameters are re-defined as a result of testing performed

3.1.18**PCB assembly uprating****CCA uprating**

uprating of a printed circuit board or circuit card assembly populated with individual components, some or all of which are operated at temperatures beyond their data sheet parameters

3.1.19**rating**

value that establishes either a limiting capability or a limiting condition for a semiconductor device

3.1.20**recommended operating conditions**

conditions for use of the component for which the component specifications, based on the component data sheet, are identified by the electronic component manufacturer

3.1.21**stress balancing**

process for thermal uprating in which at least one of the device's electrical parameters is kept below its maximum allowable limit to reduce heat generation, thereby allowing operation at a higher ambient temperature than that specified by the device manufacturer

3.1.22**target temperature range**

operating temperature range of the device in its required application

3.1.23**thermal uprating****uprating**

process to assess the capability of a part to meet the performance requirements of the application in which the device is used outside the manufacturer's specified temperature range

Note 1 to entry: Terms such as "upscreening", "retest", "up-temperature testing" and other similar variations are subsets of or encompassed by the overall uprating process.

3.1.24**wider temperature range**

target temperature range outside the manufacturer-specified temperature range

Note 1 to entry: It may include temperatures that are higher or lower than the manufacturer-specified temperature range, or both.

3.2 Abbreviated terms

ADHP	aerospace, defence and high performance
AQEC	aerospace qualified electronic component
ATP	acceptance test procedure
CAGE	commercial and government entity
CCA	circuit card assembly
COTS	commercial off-the-shelf
CMOS	complementary metal-oxide-semiconductor
ECMP	electronic components management plan
ESD	electro-static discharge
ESS	environmental stress screening
HALT	highly accelerated life testing
HAT	highly accelerated test
HASS	highly accelerated stress screening
ID	Identification
PCB	printed circuit board
PCN	process change notice
PPM	parts per million
SD	sigma deviation
QA	quality assurance

4 Selection provisions

4.1 General

Selection provisions are described below.

The flow chart of Figure 1 describes a typical approach for using devices outside the electronic component manufacturer's specified temperature range, by considering their selection (4.2), their capability assessment (4.3), their quality assurance (4.4 and 4.5) and their documentation (4.7).

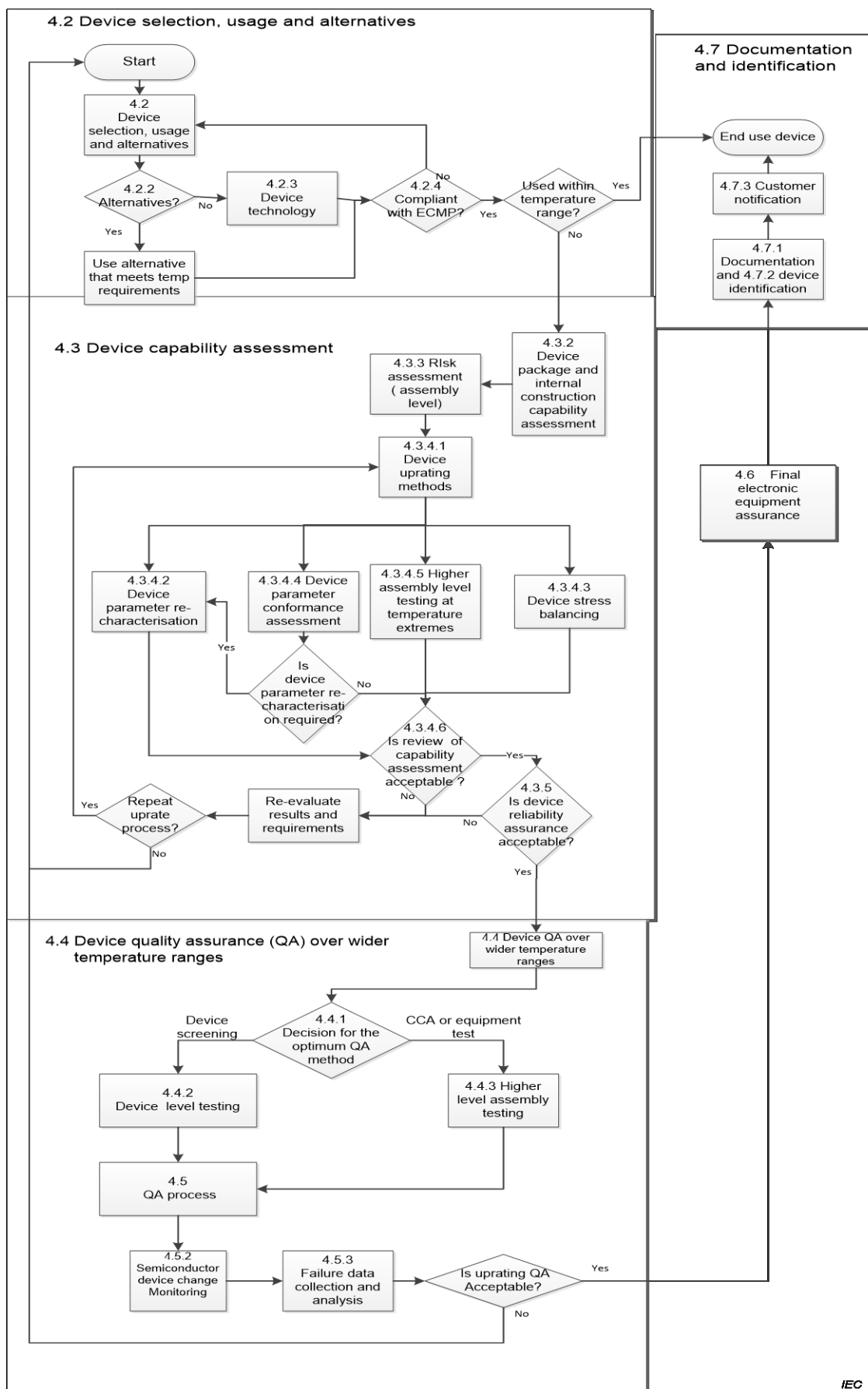
The use of devices that operate outside the temperature ranges specified by the device manufacturer is discouraged; however, such usage can occur if other options prove to be impossible, unreasonable, or impractical. Justification for such usage can be based on availability, functionality, or other relevant criteria.

Such operation is not cause for unstable part operation or loss of electronic equipment function nor is the device to be operated beyond its absolute maximum data sheet ranges (for example maximum junction temperature).

The electronic equipment manufacturer uprating the component utilizes a process to demonstrate that the component will meet reliability and lifetime requirements of the ADHP application.

Additionally, operation of the device beyond the electronic component manufacturer's limits can result normally in loss of warranty by the device manufacturer.

NOTE The headings of Clause 4 are keyed to the actions and decisions of Figure 1.



IEC

Figure 1 – Flow chart for semiconductor devices over wider temperature ranges

4.2 Device selection, usage and alternatives

4.2.1 General

The electronic equipment is designed and devices are selected so that, initially and throughout electronic equipment life, no absolute maximum value for the intended service is exceeded for any device under the worst probable operating conditions.

Operating condition examples include the following: supply voltage variation, electronic equipment device variation, electronic equipment control adjustment, load variations, signal variation, environmental conditions, and variation in characteristics of the device under consideration and of all other electronic devices in the electronic equipment.

4.2.2 Alternatives

A review of alternatives is to be performed prior to using a device outside the manufacturer's specified temperature range. If an alternative can be shown to be reasonable and practical, then it is selected. The results of this evaluation are then documented.

Examples of potential alternatives include:

- using a device specified over the required temperature range, with identical function, but from a different manufacturer;
- using a device specified over the required temperature range, with identical function, but a wider specified temperature range. Examples include: AQEC (aerospace qualified electronic components according to IEC TS 62564-1), automotive grade specified devices, or other extended range specified devices;
- using a device specified over the required temperature range, with identical function, but a different package;
- using a device specified over the required temperature range, that has slightly different specified parameter limits, but which still meets the electronic equipment design goals;
- using a device with identical function, but a specified temperature range that still meets the application requirement;
- using a device specified over the required temperature range, but with a different function, and compensating by making changes elsewhere in the electronic equipment design;
- modifying the device's local operating environment, for example, adding cooling, etc.;
- modifying the electronic equipment's specified ambient temperature requirement, in co-operation with the customer;
- modifying the electronic equipment's operating or maintenance procedures, in co-operation with the customer; and
- negotiating with the device manufacturer to provide assurance over the wider temperature range.

For most applications, the preferred device for use in a wider temperature range is the one for which the extension beyond the specified range is the least, i.e., upon making the decision to uprate a given manufacturer's part and if the manufacturer offers the device in various temperature ranges, then the widest temperature range is selected. For example, given the choice to uprate a manufacturer part available in a commercial temperature range (0 °C to 70 °C) versus the same device available in industrial grade (–40 °C to 85 °C) or automotive grade (–40 °C to 125 °C), then the device having the widest range is selected.

4.2.3 Device technology

The technology of a device and its package are to be identified and understood in sufficient detail to assess the likelihood and consequences of potential failure mechanisms. If available, manufacturer data, information and/or guidance are collected at the onset.

4.2.4 Compliance with the electronic component management plan

All devices considered for use in wider temperature ranges are to be compliant with the electronic equipment manufacturer's ECMP.

NOTE IEC TS 62239-1 is a resource for an ECMP.

4.3 Device capability assessment

4.3.1 General

The assessment of device capability needs to assure that not only are device parameters acceptable, but also that device functionality and functionality of the related circuit application are acceptable as well. Therefore, functional testing at the application or higher levels is recommended.

4.3.2 Device package and internal construction capability assessment

Device qualification test data and other applicable data when available are to be analysed to assure that:

- a) they support the operation of the device over the end use temperature range and that the package and internal construction type used in device qualification is the same as that to be used in the end application;
- b) the package and internal construction can withstand the stresses resulting from wider temperature cycling ranges, and that the package materials do not undergo deleterious phase changes or changes in material properties in the wider temperatures.

If data are not available, then relevant testing based on the application is to be considered.

4.3.3 Risk assessment (assembly level)

A preliminary risk assessment is to be performed to help guide decisions regarding the method(s) of capability assessment and quality assurance (QA) to be used, as well as how and when they are applied. Understanding the risks on an application-specific basis enables "risk informed" decision-making and thereby a prediction of the impact of critical decisions. The process for assessing risks considers applicable factors associated with the use of devices beyond the manufacturer's specified temperature range. Risk factors in this assessment may include:

- application criticality into which the device will be used;
- consequences of failure at device, circuit assembly and system level;
- type or technology of the device under consideration;
- manufacturer data available for the device;
- quality/reliability monitors employed by the manufacturer including lot-to-lot variation;
- comprehensiveness of production assembly-level screens performed at extended temperature;
- identification of both managed and unmanaged risks.

Details about the likelihood of occurrence, consequences of occurrence, and acceptable mitigation approaches for each identified risk are generated. Each risk normally falls into one of the following categories:

- functionality risks: risks for which the consequences of occurrence are loss of electronic equipment, loss of mission, or unacceptable performance. Functionality risks impair the product's capability to operate to the customer's specification;
- "produceability" risks: risks for which the consequences of occurrence are schedule impacts. "Produceability" risks determine the probability of successfully

manufacturing/fabricating the product (where “successfully” refers to some combination of schedule, manufacturing yield, quantity and other factors).

Several approaches are possible, and each approach constitutes a unique mixture of risk mitigation factors. The results of a preliminary risk assessment should provide insight and assistance to the selection of a viable approach or approaches for establishing the capability of devices being used outside the manufacturer’s specified temperature range.

NOTE Up-rating can be supported by exchanging technical information with the original device manufacturer.

4.3.4 Device uprating methods

4.3.4.1 General

Devices are to be reviewed to determine the optimum method of uprating based on risk assessment. Options include:

- a) device parameter re-characterisation, see 4.3.4.2;
- b) device stress balancing, see 4.3.4.3;
- c) device parameter conformance assessment, see 4.3.4.4;
- d) higher assembly level testing, see 4.3.4.5.

Where possible, devices for uprating are taken from a single lot. The use of additional lots (or samples) may be utilized to undergo testing as part of the initial characterisation if it is determined that lot variations may exist.

4.3.4.2 Device parameter re-characterisation

Device parameter re-characterisation consists of characterising the device parameters over a temperature range beyond that specified by the device manufacturer and, as a result, re-specifying the data sheet parameters targeted for uprated values or tolerances in the wider temperature range. The device then may be used in applications in which the newly specified parameters provide the required functionality. To effectively assess device manufacturing variability, multiple date codes need to be considered, with the recognition that this can be application- and usage-rate dependent.

If device parameter re-characterisation is chosen for capability assessment, then the process described in Annex A is followed and is used in conjunction with a quality assurance process as described in 4.4.

4.3.4.3 Device stress balancing

Device stress balancing consists of operating the device at an ambient temperature above that specified by the device manufacturer and compensating by reducing at least one of the other operating parameters, for example, power or speed, to the extent that the junction temperature remains below its maximum rating, with an acceptable specified margin.

If device stress balancing is chosen for capability assessment, then the process described in Annex B is followed.

4.3.4.4 Device parameter conformance assessment

If device parameter conformance is chosen for capability assessment, then the devices are tested over the entire wider temperature range using the original specification parameters, according to the process described in Annex C.

Sampling procedures and failure criteria for device testing are according to Annex C. Where fewer than 100 % are sampled, then device testing also includes testing at a higher level of assembly over the entire wider temperature range.

If the results of the parameter conformance assessment testing show that the parameters do not fall within the original specification parameters, the capability assessment can continue by performing a parameter re-characterisation according to 4.3.4.2.

4.3.4.5 Higher assembly level testing at temperature extremes

Higher assembly level testing at temperature extremes consists of testing the device over the entire wider assembly ambient temperature range, while the device is incorporated into a higher level of assembly.

This method applies to one device type in one or multiple locations or several device types candidates for uprating in a same assembly.

If higher assembly level testing is chosen for capability assessment, then the process described in Annex D is followed.

NOTE 1 A higher level of assembly can include a module, a printed circuit card, another sub-assembly, or the end item.

NOTE 2 The intent of 4.3.4.4 and 4.3.4.5 is to ensure that, if testing is used to assess device capability, then each device is tested at least once over its entire wider operating ambient temperature range.

NOTE 3 Testing can employ HAT, HALT or other similar testing to address higher level requirements.

Higher assembly level testing results are applicable only to the current design revision of the assembly. For further assembly revisions, additional testing or analysis should be performed.

The following steps are followed:

- a) Perform a circuit element functional mode analysis for each location of the device candidate for uprating to determine the device functions/parameters to be tested in order to assure assembly functionality across the target ambient temperature range.
- b) Review the assembly level test plan to determine its capability to test the parameters required for successful operation in the assembly. If the test plan is not capable, and cannot be modified to be capable, then this method of uprating is rejected for the application.
- c) Conduct the test, analyse the results, and document the conclusions.
- d) Insert instructions in the maintenance procedures to require full acceptance test over the target ambient temperature range. This testing applies after every maintenance action that involves replacement of an electronic device at the assembly level for which the original capability assessment was performed, unless the maintenance manual provides adequate alternate procedures. This test should be conducted at an assembly level at which the original capability assessment was done, or higher.

4.3.4.6 Review of capability assessment

A review of the results of the capability assessment is conducted. Based upon the acceptability of the results, a decision is made to proceed.

4.3.5 Device reliability assurance

Device manufacturers generally qualify devices (including reliability assessment) using the same processes, regardless of the temperature ranges for which they are specified. Generally, they do not represent their products as having a guarantee of lifetime in any application, because they do not know what the use conditions will be. Caution is exercised when using past experience of the device within the manufacturer's specified temperature range to infer reliability outside of the manufacturer's specified temperature range.

The application of each device and any related impacts on reliability should be assessed. New and/or accelerated failure mechanisms, which might be evident at the wider temperature

range, are to be clearly identified and their effects on reliability established. If deemed necessary, additional testing can be implemented to address application reliability concerns.

The distribution of time at which a device is actually operating beyond a device manufacturer's specified temperature range and the related impact on reliability need to be considered.

NOTE 1 Upgrading conditions often occur only as "corner conditions" or for specified extreme environments which are seldom experienced.

The following steps apply:

- a) Qualify the devices according to the requirements of the user's electronic component management plan, as specified in 4.2.4. Also, qualify electrical performance of the devices over the intended range of the application operating and environmental conditions. Determine a temperature margin, supported by analysis using adequate data from the intended application, between the maximum operating junction temperature and the absolute maximum rated junction temperature.
- b) The absolute maximum rating of the junction temperature of the device is as defined in 3.1.12, where a default margin of 20 °C below the absolute maximum junction temperature is considered to be best practice. Other margins may be used if the device user has data to justify them.

If the junction temperature average, T_j , of the device is expected to approach the maximum in the application, the reliability impact is to be addressed.

NOTE 2 Device reliability can decrease as junction temperature, T_j , approaches the maximum. This is a function of time in application at that temperature.

NOTE 3 Many avionics applications specify a high temperature environment in which the device is required to operate. Thermal conditions which are rarely experienced do not significantly affect device reliability assurance in wider temperature ranges.

4.4 Device quality assurance (QA) over wider temperature ranges

4.4.1 Decision for the optimum QA method

Regardless of the process used to assure device capability, the quality assurance (QA) processes documented in the electronic equipment manufacturer's ECMP are applied to the device.

Options for QA include:

- 1) performing testing at the individual component level,
- 2) performing testing at the higher level assembly (for example, at CCA level or electronic equipment) level,
- 3) a combination of 1), 2).

All relevant data resulting from the capability assessment are analysed along with other factors to determine the optimum quality assurance process methodology. The decision for the optimum QA and methodology considers mitigation of risk to the electronic equipment including possible impacts to the supply chain. Trade studies are useful to review potential impacts of each option taking into account, for example that:

- testing performed at the component level can cause increased errors due to additional process steps with, for example, additional handling, which can introduce potential issues such as electronic component terminations damage and/or ESD damage, etc. If this operation is sub-contracted, possible negative yields, additional lead time, and/or cost impacts can occur,

- testing performed at the electronic equipment or CCA level could introduce possible yield effects and/or the need for additional manufacturing rework which could impact schedule and/or cost.

4.4.2 Device level testing

Parametric testing at the device level is performed at the uprated temperature range and covers those parameters used outside of the electronic component manufacturer's data sheet specifications.

Based on data derived from such testing, testing can be reduced or eliminated by satisfactory test history and by effective supplier change notice monitoring. The sampling rate, confidence limits, and decision criteria are as stated in Annex C.

4.4.3 Higher level assembly testing

Assembly level tests are designed to test basic functional performance of the assembly at the application temperature. The assembly testing consists typically of either testing at the electronic equipment level or at the CCA level. Typically, all functions or "key characteristics" of the end product are verified at the sub-assembly or end-item level. The difference between the typical case and the process described here is that the device's role in these functions, or "key characteristics", of the assembly are traced, and its capability verified by assembly test over the target temperature range of the application.

4.5 QA process

4.5.1 General

The overall process is reviewed to ensure conformance to IEC TS 62239-1 ECMP including requirements for documentation, change control, failure analysis, and device identification.

4.5.2 Semiconductor device change monitoring

Device data (such as product change notices or manufacturer data) are monitored to give warning of device changes that may affect the capability of the device to operate over the wider temperature range as established in 4.3. Significant changes (for example, die change, die shrink, etc.) are evaluated and if determined to impact the uprating performance, then the entire uprating process is re-evaluated and repeated based upon the changed device.

The approach for monitoring the device design and manufacturing process change data is no different than the related requirement in the IEC TS 62239-1 ECMP specification.

An alternative option which precludes the need for change monitoring, is to procure on an initial basis all devices needed for production that have the same part configuration and are from the same lot. This assures all parts are homogeneous.

4.5.3 Failure data collection and analysis

Failure data are collected for all uprated devices. When clear trends are evident, the data are analysed, root cause identified and corrective action taken.

Failures of devices used in wider temperature ranges are to be analysed to establish the root cause of the failure.

When failure analysis is conducted, the results are to be documented.

4.6 Final electronic equipment assurance

Final electronic equipment assurance testing is performed utilizing system level tests such as ESS, HASS, ATP, etc. This testing may be performed during the higher level assembly testing step (see 4.4.3) if applied to the electronic equipment level.

4.7 Documentation and identification

4.7.1 Documentation

For each instance of device usage outside the manufacturer's specified temperature range, relevant data are documented and stored in a controlled, retrievable format.

The documented information consists of:

- electronic equipment in which the device is used;
- device identification;
- required operating temperature range;
- manufacturer-specified operating temperature of the device;
- alternatives considered and rejected;
- process for assuring device capability in the wider temperature range (including test and analysis results);
- required signatures;
- risk assessment results.

NOTE Required signatures include those of the responsible authorities within the equipment designer's organisation and, if required, those of the customers.

4.7.2 Device identification

All device identification processes are to be consistent with other industry processes.

For each instance in which a device has been determined as having met the application's wider temperature range requirements, through parameter re-characterisation (4.3.4.2), device stress balancing (4.3.4.3) or device testing (4.3.4.5), the device's status is identified as having met the requirements specified in the design activity's uprating specification. The identification requirements are as specified in the design activity's uprating specification and include the design activity's unique identifier such as the CAGE code, logo, or acronym and the part number assigned by the design activity. For each occurrence of uprating, the parts are to be separately identified as meeting the requirements of the application relative to the same parts which were not uprated. The method of identification enables all relevant activities such as spares and maintenance to establish that the device has met the requirements of parameter re-characterisation (4.3.4.2), device stress balancing (4.3.4.3) or device testing (4.3.4.5).

If the device is marked, then marking is in addition to the existing/original manufacturer's marking and is to be readable when the device is mounted in its application. All markings applied are to be permanent and legible. When the device is marked, then the marking details are to appear in the uprating process documentation.

4.7.3 Customer notification

The form of Figure 2 is proposed for use in documenting semiconductor device usage in wider temperature ranges.

WIDER TEMPERATURE RANGE DEVICE USAGE REPORT	
Equipment name (if applicable) _____	Date _____
Equipment ID number (if applicable) _____	Program manager _____
Name _____	Device engineer _____
ID no. before uprate _____	ID no. after uprate _____
Equipment required temperature range:	Min _____ Max _____ (ambient)
Manufacturer's specified device temperature range:	Min _____ Max _____ (ambient)
Is the device compliant with the equipment manufacturer's electronic component management plan? Yes _____ No _____	
What alternative solutions were evaluated, and why were they rejected?	
Capability assessment process: Is the package capable? _____	Device parameter re-characterisation _____ Device stress balancing _____ Assembly testing _____ Device testing _____
Is the device capable of operating in the required temperature range without significantly increased risk of catastrophic failure, unstable operation, loss of equipment function, or adversely affecting the application-specific reliability of the device? (Reference or attach capability assessment report)	
Quality assurance process:	Sample plan and monitoring _____ 100 % equipment test _____ 100 % device test _____
Is the device's quality assured? (Reference or attach QA plan) Device marking.... [yes] [no]; if yes: part marking is as follows: _____	
Approvals _____	

To be used with the chosen uprating method report form(s) and table:

- Annex A, Figure A.8, for report form related to device parameter re-characterisation method and table based on the template of Table A.2, for parameter re-characterisation registration;
- Annex B, Figure B.6, for report form related to stress balancing method;
- Annex C, Figure C.4, for report form related to parameter conformance testing method;
- Annex D, Figure D.2, for report form related to higher level assembly test at temperature extremes method.

Figure 2 – Report form for documenting device usage over wider temperature ranges

Annex A (informative)

Device parameter re-characterisation

A.1 Glossary of symbols

T_{room} :	Room temperature (25 °C)
$T_{\text{rated-max}}$:	Maximum temperature at which the part manufacturer guarantees operation of a part in accordance with the published data sheet
$T_{\text{rated-min}}$:	Minimum temperature at which the part manufacturer guarantees operation of a part in accordance with the published data sheet
$T_{\text{req-max}}$:	Maximum temperature at which the part is required to operate in a system
$T_{\text{req-min}}$:	Minimum temperature at which the part is required to operate in a system
$T_{\text{test-max}}$:	Maximum temperature at which the part is tested, usually more than $T_{\text{req-max}}$
$T_{\text{test-min}}$:	Minimum temperature at which the part is tested, usually less than $T_{\text{req-min}}$
UL :	Maximum limit of a parameter value in the part manufacturer's specified temperature range (specified by the part manufacturer)
LL :	Minimum limit of a parameter value in the part manufacturer's specified temperature range (specified by the part manufacturer)
UL_{Max} :	Maximum allowable upper limit for a parameter for proper system operation
LL_{Min} :	Minimum allowable upper limit for a parameter for proper system operation
UL_{New} :	New maximum parameter value limit if the parameter limit is modified
LL_{New} :	New minimum parameter value limit if the parameter limit is modified
M_{UL} :	Margin of tested parameter value at extremes of target application temperature range with UL
M_{LL} :	Margin of tested parameter value at extremes of target application temperature range with LL
$M_{UL\text{-req}}$:	Required margin of tested parameter value at extremes of target application temperature range with parameter limit
$M_{LL\text{-req}}$:	Required margin of tested parameter value at extremes of target application temperature range with parameter limit
E :	Precision of sampling for mean of a parameter
E_A :	Measurement inaccuracy
σ :	Population standard deviation
s :	Sample standard deviation
n :	Multiplier for standard deviation (typically 3)
μ :	Mean of a population
Part manufacturer-specified temperature range:	The operating temperature range over which the component specifications, based on the component data sheet, are guaranteed by the component manufacturer. The range between $T_{\text{rated-min}}$ and $T_{\text{rated-max}}$

Target temperature range:	The operating temperature range of the part in its required application. This temperature range may be wider than the part manufacturer's specified temperature range.
	The range between $T_{\text{req-min}}$ and $T_{\text{req-max}}$
Test temperature range:	The temperature range over which a part is tested for parameter re-characterisation.
	The range between $T_{\text{test-min}}$ and $T_{\text{test-max}}$

Formula (A.1) gives the relationship of the above temperatures:

$$\begin{aligned} \text{a) } T_{\text{test-min}} &\leq T_{\text{req-min}} \leq T_{\text{rated-min}} \leq T_{\text{room}} \\ \text{b) } T_{\text{room}} &\leq T_{\text{rated-max}} \leq T_{\text{req-max}} \leq T_{\text{test-max}} \end{aligned} \quad (\text{A.1})$$

It should be noted that for the conditions expressed in Formula (A.1) above, parameter re-characterisation can apply for either condition a), condition b), or both.

A.2 Rationale for parameter re-characterisation

A.2.1 General

The re-characterisation process is conducted by the user of the device, or a designated test facility. It measures electrical parameters and their variations over a target temperature range that is wider than that specified by the device manufacturer. Based on test results, the data sheet parameters may be used as published, or modified if test results so indicate. It can be necessary to maintain the data sheet parameters for some target temperature ranges, and to modify others. The new parameter limits may not be applicable to all applications. Figure A.1 illustrates the rationale for parameter re-characterisation.

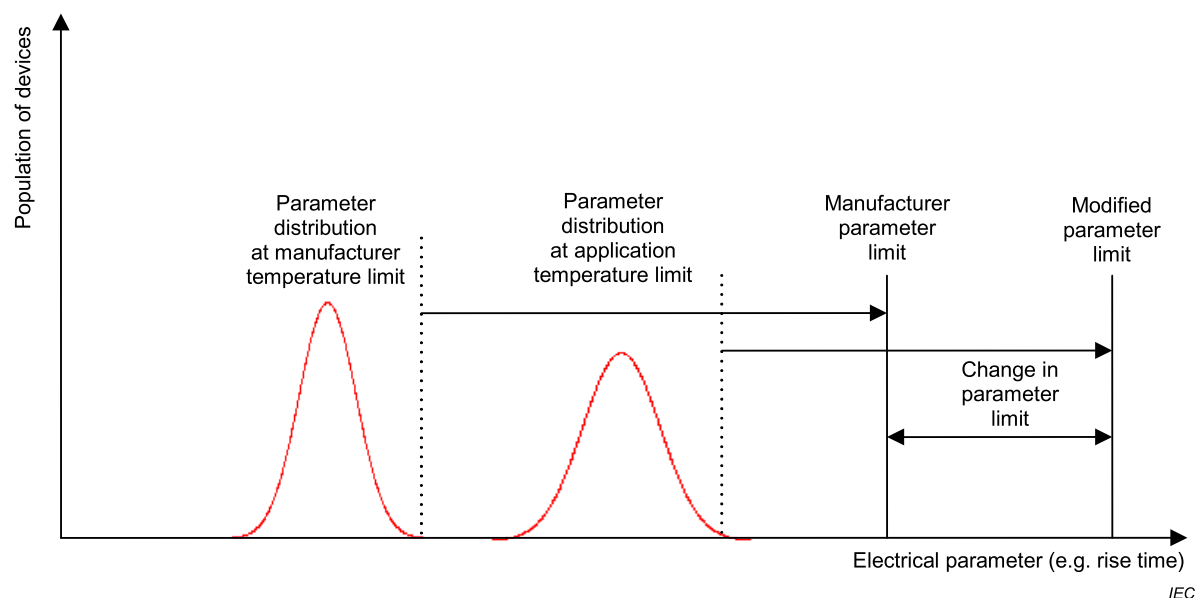


Figure A.1 – Parameter re-characterisation

A.2.2 Assessment for uprateability

Before any testing is performed, data from all available sources are analysed to determine if it is reasonable to attempt parameter re-characterisation. Typical sources of such data include

users of similar devices in similar applications, test laboratories, manufacturers, and industry organisations.

Device manufacturers' processes assure the quality of devices within their specified temperature limits. If they can be obtained from the device manufacturer, data from these processes can provide insight about a device's expected performance over the target temperature range. Simulation models, for example, the BSIM3 model for short channel MOSFETs, can be used to estimate the effects of temperature variation on device parameters, and therefore can be used to assess their uprateability, see [1]¹.

A.3 Capability assurance

A.3.1 Description

Parameter re-characterisation is a method of thermal uprating in which the part parameters are characterised over the target temperature range, using processes similar to those used by the device manufacturer for original device characterisation. If parameter re-characterisation is successful, the device may be used in applications in which the re-characterised parameters are acceptable.

A.3.2 Parameter re-characterisation process

A.3.2.1 General

Figure A.2 shows a flow diagram of the parameter re-characterisation process.

A.3.2.2 Critical parameter selection

All electrical parameters that are critical to the application are identified and re-characterised over the entire target temperature range of the application. Possible interdependence of datasheet parameters, for example logic voltage dependence on supply voltage, is considered in deciding which parameters to include.

A.3.2.3 Sample size determination

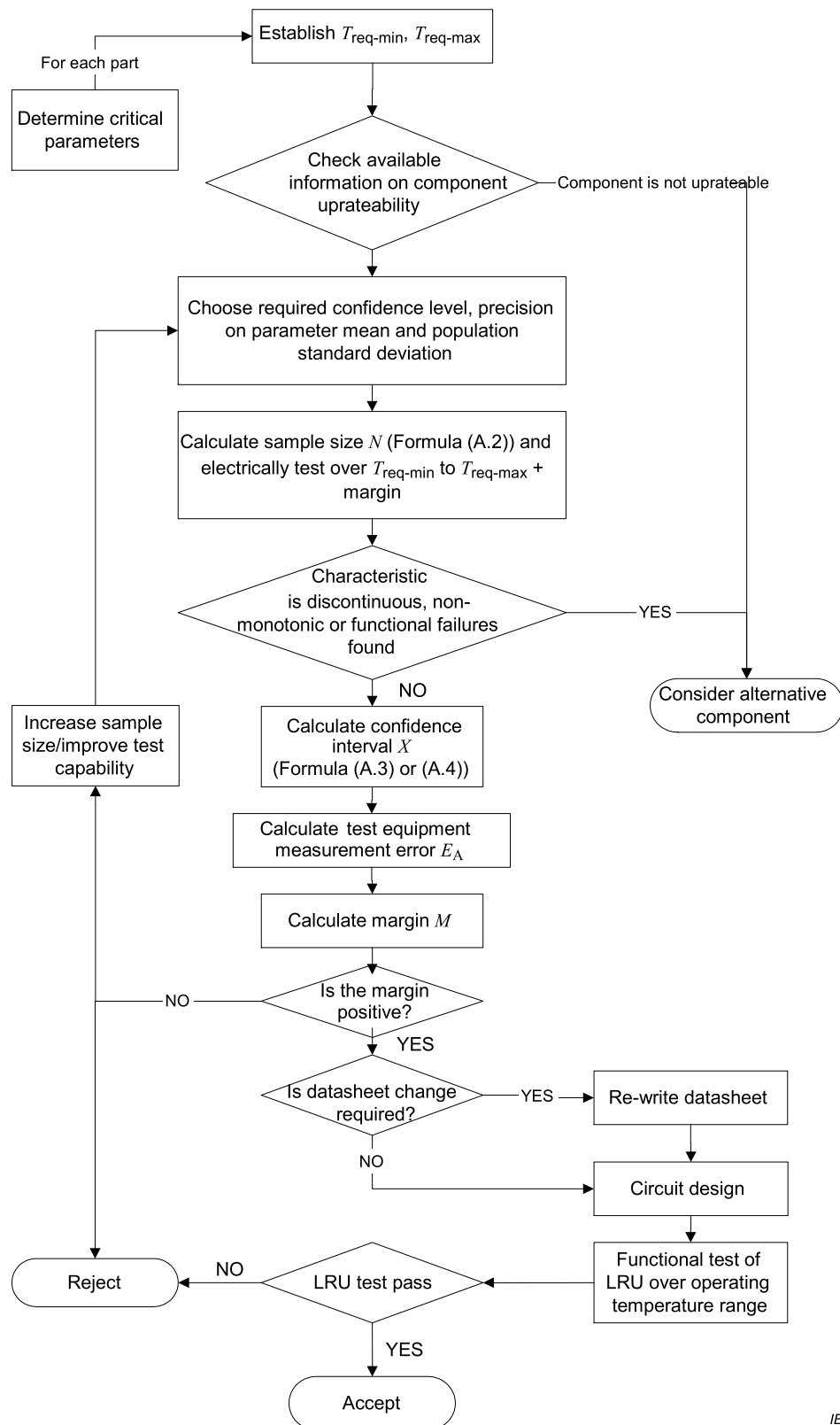
Sample sizes for parameter re-characterisation are to be sufficiently large to provide reasonable assurance that normal variations in the re-characterised parameters will not cause the parameters to be outside their re-characterised limits. The sample size is determined for each instance of parameter re-characterisation. To effectively assess device manufacturing variability, multiple date codes need to be considered, with the recognition that this can be application- and usage-rate dependent. Factors to be considered may include:

- number of devices available for testing,
- types of parameters to be tested,
- target temperature,
- resources required to conduct the tests,
- desired confidence level for the results,
- desired parameter margins, and
- other factors relevant to the device and the application.

For each instance of parameter re-characterisation, the following information should be included in the uprating documentation:

¹ Numbers in square brackets refer to the Bibliography.

- process used to determine the sample size,
- statistical distribution (assumed or known) of the parameters,
- confidence level, and
- other relevant information.



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Figure A.2 – Flow diagram of parameter re-characterisation capability assurance process

In most instances, a normal distribution is assumed. For the normal distribution, the sample size, N , is defined in references [2] [3]:

$$N = \left[\frac{Z_{\alpha/2} \times \sigma}{E} \right]^2 \quad (\text{A.2})$$

where

E is the required precision on the parameter mean,

$Z_{\alpha/2}$ is the value of the standard normal variable at confidence level $(1 - \sigma) \times 100 \%$, and

σ is the standard deviation of the population.

In Formula (A.2), the sample mean is within $\pm E$ of the true mean, with a probability of $(1 - \sigma)$. Table A.1 shows the results of an example calculation (standard deviation values were obtained from typical device data).

Table A.1 – Example of sample size calculation

	Precision (E)	Standard deviation (σ)	Sample size		
			$\sigma = 90 \%$	$\sigma = 95 \%$	$\sigma = 99 \%$
Propagation delay	0,2 ns	0,3 ns	7	9	15
Input current	0,005 μA	0,006 7 μA	5	7	12
Supply current	0,15 μA	0,2 μA	5	7	12

NOTE 1 If it is known that another distribution fits the data, then appropriate statistics are used.

NOTE 2 To be completely rigorous, each parameter at each test temperature has its own specific distribution. In practice, this degree of detailed knowledge is rarely available, unless there is evidence to the contrary, and it is acceptable to assume that the same distribution parameters apply to all electrical test parameters at all test temperatures.

For large sample sizes, i.e., greater than thirty, the confidence interval estimate of the mean is:

$$\bar{X} \pm \left(Z_{1-\alpha/2} \times \frac{\sigma}{\sqrt{N}} \right) \quad (\text{A.3})$$

For small sample sizes, i.e., less than thirty, then the Student's t distribution should be used, and the confidence interval estimate of the mean is:

$$\bar{X} \pm \left(t_{1-\alpha/2} \times \frac{s}{\sqrt{N}} \right) \quad (\text{A.4})$$

where s is the sample standard deviation and $t_{\alpha/2;N-1}$ is the value of the Student's t distribution at the confidence level $(1 - \alpha) \times 100 \%$ and $N - 1$ degrees of freedom.

A.3.2.4 Testing for re-characterisation

Parameter re-characterisation tests are conducted over the entire target temperature range, and also consider temperature margins above the maximum and below the minimum target temperatures. During the parameter re-characterisation process, the device absolute temperature limits are known and understood, and if they exceed any of these absolute limits

during this process they are controlled and performed only to provide additional understanding of device behaviour. Devices used in actual applications are not to exceed absolute maximum ratings.

Tests are conducted at various temperatures within the target temperature range. The number of test temperatures, and the intervals between them, may not be the same for all instances of parameter re-characterisation. Factors to be considered in determining the test temperatures may include:

- device manufacturer's specified temperature range;
- other thermal data obtained from the device manufacturer, for example, thermal conductivity, etc.;
- target temperature range;
- other uses of the test data, for example, performance derating; and
- previous relevant experience with the device.

Additional test temperatures may be specified on the basis of tests conducted during parameter re-characterisation. For example, if a plot of a given parameter versus temperature indicates the relationship may not be linear, additional tests are performed to determine its exact nature.

A device can satisfy its parameter specifications, but still fail to function in an application. Therefore, functional testing at the application or higher levels is considered. For digital devices, gate level design information is required to develop software to achieve specific fault coverage. If the full set of test vectors is not available, the percent fault coverage is difficult to determine without detailed knowledge of device architecture. Again, functional testing in the application is to be considered.

Testing may be performed in-house or at an external test house. In either case, the equipment supplier is responsible for the tests and their results.

Prior to parameter re-characterisation testing, a set of requirements and limitations on the electrical parameters is developed. The requirements and limitations depend on the application.

Acceptable upper (M_{UL-req}) and lower (M_{LL-req}) margin limits should be established for each modified parameter.

A.3.2.5 Assessment of electrical test results

If the test results indicate that there are no functional failures, if no discontinuities are observed in any of the parameter versus temperature plots, and if the modified parameter limits are acceptable for the application, then the uprating process can be considered successful.

A.3.2.6 Re-characterised parameter value calculation

A.3.2.6.1 General

Re-characterised parameter values include both the nominal values and their limits. The limits are determined by combining variations due to sampling, parameter values, and test equipment accuracy to the nominal values. Figure A.4 illustrates the method by which they are combined.

A.3.2.6.2 Nominal values

The nominal value of a re-characterised electrical parameter is the value selected for use in designing equipment with the re-characterised device. It can be constant over the target

temperature range, or it may vary with temperature in a predicted manner. Usually, the mean value of the test results for a given parameter at a given temperature is designated as the nominal value, although other values may be chosen if warranted.

A.3.2.6.3 Variation

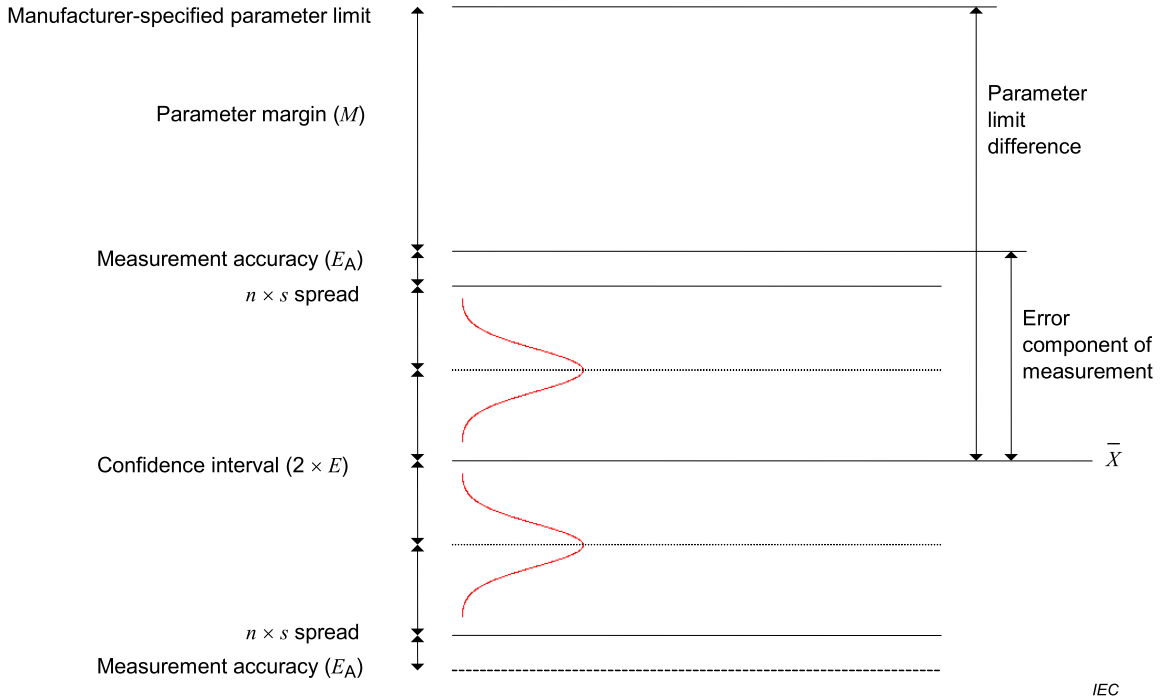


Figure A.3 – Margin in electrical parameter measurement based on the results of the sample test

Variation due to sampling is the confidence interval described in A.3.2.3. It is shown as $2 \times E$ in Figure A.3.

Parameter variation is shown as $n \times s$ in Figure A.3. Usually, the standard deviation of the test sample is used as the measure of parameter variation, with the number of standard deviations, n , determined on the basis of acceptable risk. Variation in test equipment accuracy is shown as E_A in Figure A.3. Test equipment accuracy is calculated with standard methods found in basic statistics texts. It can vary according to the test temperature.

A.3.2.6.4 Margin calculation

The parameter margin, M , is calculated by:

$$M = UL - \bar{X} - E_A - n \times s - E \quad (\text{A.5})$$

A.3.2.7 Parameter limit modification

If a given parameter margin is considered inadequate ($M < 0$), then the data sheet parameter limits may be modified to provide new limits to be used in equipment design. Parameter limit modification begins with the selection of the required margin for a given temperature. Potential variations calculated in A.3.2.6.2, A.3.2.6.3 and A.3.2.6.4 are added to, or subtracted from, the nominal value of the parameter at the given temperature. If the modified parameter values thus obtained are beyond the maximum and minimum parameter limits determined in A.3.2.6, then the device is not uprateable through parameter re-characterisation. Figure A.4 shows an example of the parameter limit modification process. In this example, the

new parameter limit is below the maximum allowable parameter limit, and thus acceptable. Table A.2 shows an example of re-characterising a 0 °C to 70 °C rated part to a –55 °C to 125 °C part.

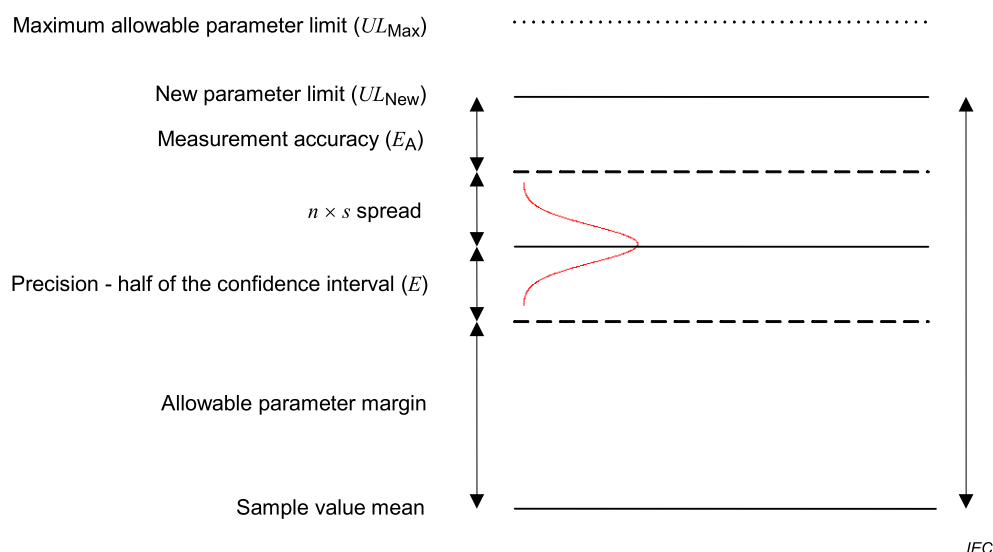


Figure A.4 – Schematic diagram of parameter limit modifications

**Table A.2 – Parameter re-characterisation example:
SN74ALS244 octal buffer/driver**

Parameter		Commercial limit	Military limit	Measured value at military limit	Derated limit (calculated) ^a
t_{PLH} (ns)	Min.	2,0	1,0	5,1	1,8
	Max.	10,0	16,0	12,8	15,2
t_{PHL} (ns)	Min.	3,0	3,0	6,7	1,9
	Max.	10,0	12,0	10,2	11,1
V_{OH} (V)	Min.	3,50	3,50	3,75	3,31
V_{OL} (V)	Max.	0,40	0,40	0,18	0,42
I_{CCH} (mA)	Min.	9,00	9,00	9,10	7,65
	Max.	17,00	18,00	14,14	18,60
I_{CCL} (mA)	Min.	15,00	15,00	14,71	14,50
	Max.	24,00	25,00	19,36	26,00

^a Assumes the same degree of errors and standard deviation at all temperatures. The margins at the commercial temperature limit are maintained at the military temperature limit.

Table A.2 can be used as a template for registering the parameters of device re-characterisation.

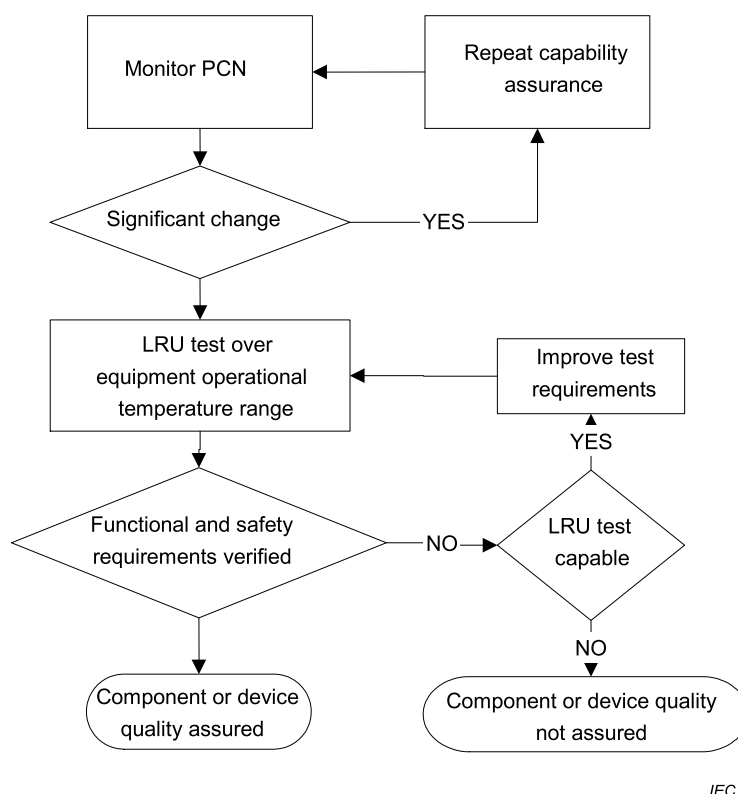
A.3.3 Application capability assessment

A representative sample of the assembly containing the devices that have been uprated by parameter re-characterisation is tested to verify that they will perform their intended function. The uprating process can be considered successful only if the higher level assembly performs properly.

A.4 Quality assurance

The ongoing quality of successfully uprated devices is by monitoring the device process change notices (PCNs) obtainable from the device manufacturer or distributor and by equipment level tests over the target temperature range, plus (or minus) a margin². Functional testing should be sufficiently rigorous to verify all system functional requirements.

Figure A.5 illustrates the flow chart for device quality assurance.



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Figure A.5 – Parameter re-characterisation device quality assurance

A.5 Factors to be considered in parameter re-characterisation

Data used in initial uprateability assessments may not be an accurate indicator of expected future performance.

Simulation models are used with caution. When they are made available to the public, they are often "sanitized" to mask proprietary information, and thus may not be accurate indicators of the analog behaviour of devices.[1] The uprateability assessment process is used only to eliminate unpromising candidate devices, and not as a substitute for electrical testing.

Data sheets do not always list all electrical parameters. This is especially true for degradation type parameters, for example gate current, substrate current, trigger currents for latchup, etc. These parameters may not be important in manufacturer-specified temperature ranges, but could be significant at target temperature ranges. When the manufacturers' test procedures are not available, it is difficult to measure these parameters, and they should be estimated. If

² To account for system variations, it can be advisable to test the system beyond its specified temperature limits. However, operating a system beyond its temperature specifications can overstress other components of the system besides the candidate part and result in invalid failures.

the initial assessment indicates that any such parameters could be of concern at the target temperature, then the revised datasheet should include limiting values for these parameters.

Some device lots can include outliers (see [5]), which limits the efficacy of sample testing (see Figure A.6).

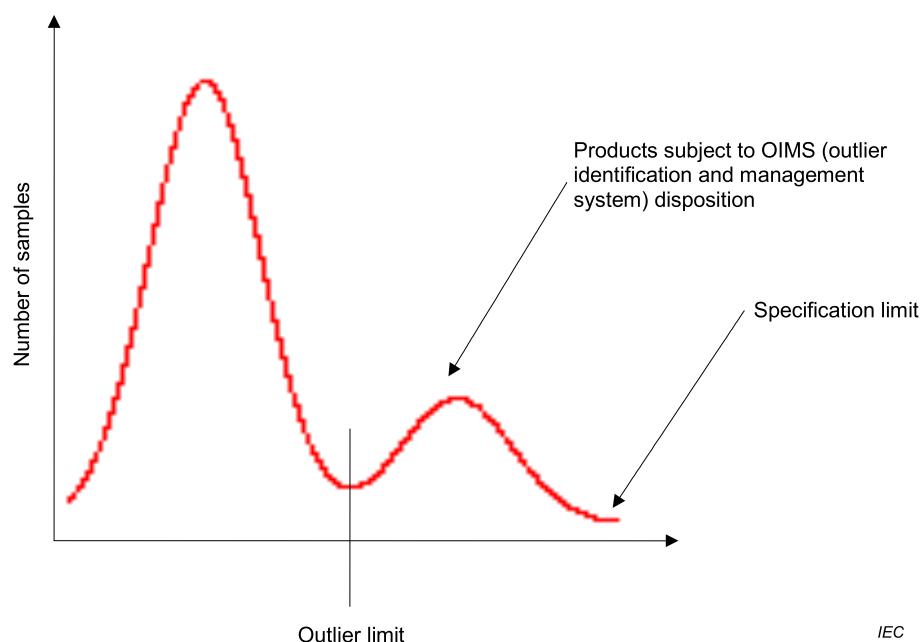


Figure A.6 – Schematic of outlier products that can invalidate sample testing

If the test temperature range does not extend beyond the target temperature range, it is difficult to detect discontinuities in the parameter versus temperature curves. Likewise, if the test temperature intervals are too wide, discontinuities within the test temperature range can be missed. If the test results indicate non-monotonic behaviour, then additional temperature points may have to be added. Figure A.7 shows an example of an intermediate peak of an electrical parameter (see [5]).

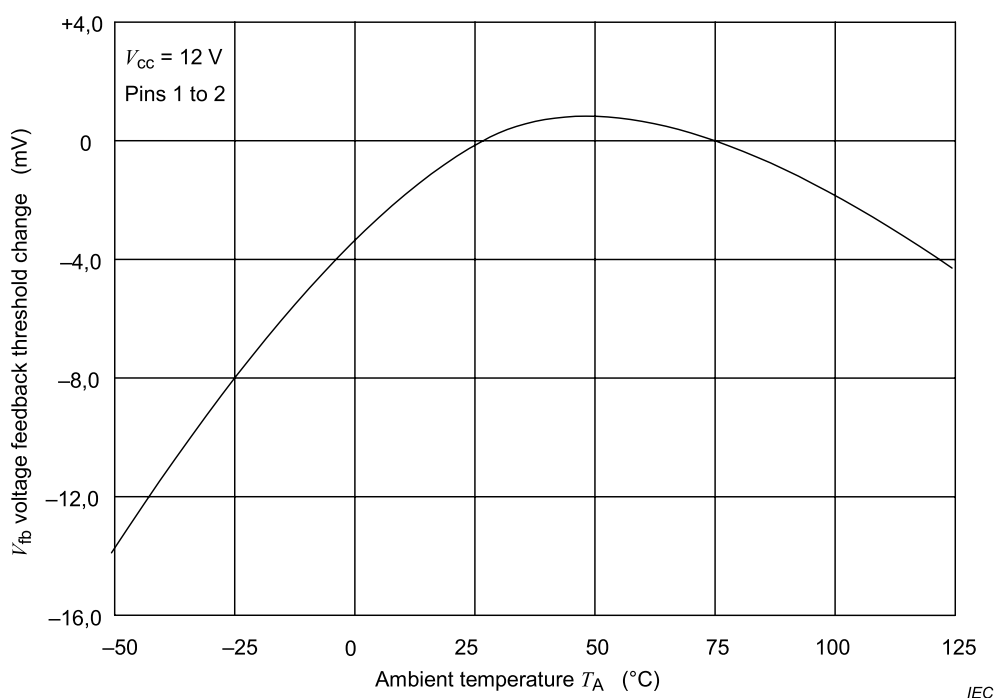


Figure A.7 – Example of intermediate peak of an electrical parameter: Voltage feedback input threshold change for Motorola MC34261 power factor controller³, see [4]

During initial characterisation, it is also necessary to check for hysteresis in electrical test data, but hysteresis tests can be eliminated if the testing does not reveal its existence. Hysteresis can be observed during temperature characterisation because (a) part characteristics change due to exposure to high or low temperature, or (b) thermal equilibrium is not reached. If hysteresis effects are observed, then dwell times at temperature should be increased. If that is not successful, then other damage possibilities should be investigated. If no other possible reason for failure is found, then the device may not be a candidate for uprating.

Inflection points observed in electrical parameters observed at extreme temperature ranges should always be tested to determine if the failure type is hard or soft. Also, it should be determined whether the failures are due to changes in device characteristics, which could result from device failure, or from the effects of extreme temperatures on testing fixtures and equipment.

A.6 Report form for documenting device parameter re-characterisation

Figure A.8 proposes a report form for documenting device parameter re-characterisation.

³ Motorola MC34261 power factor controller is the trade name of a product supplied by Motorola. This information is given for the convenience of users of this document and does not constitute an endorsement by IEC of the product named. Equivalent products may be used if they can be shown to lead to the same results.

DEVICE PARAMETER RE-CHARACTERISATION REPORT		
Device description:		Equipment name:
Device manufacturer:		Equipment part number:
Device manufacturer part number:		Program manager:
Equipment manufacturer drawing number:		Date:
1.	Rated temperature range (case or ambient):	
2.	Usage temperature range (case or ambient):	
3.	Test specification number(s): Parametric: Functional:	
4.	Device date code(s):	
5.	#Sample size:	
6.	#Functional test passed:	
7.	#Functional test failed:	
8.	Test date:	
9.	Test results report:	
10.	Approvals:	

Figure A.8 – Report form for documenting device parameter re-characterisation

Annex B (informative)

Stress balancing

B.1 General

Stress balancing takes advantage of the power-temperature trade-off opportunity in a given application. It requires less testing than parameter conformance assessment and parameter re-characterisation, since testing is done only to confirm analytical results in the specific application. See Clause 3 for the definition of stress balancing.

B.2 Glossary of symbols

T_A :	Ambient temperature
T_{A-Max} :	Manufacturer-specified maximum ambient temperature
T_J :	Junction temperature
T_{Up-Max} :	Maximum temperature up to which the device can be uprated
T_{App} :	Ambient temperature limit required for the application
T_M :	The margin by which the Iso- T_J curve is derated
ΔT_A :	Change in the ambient temperature
P :	Power dissipation
P_{Min} :	Minimum power dissipation at which the device can be operated in the system, calculated from maximum allowable limits on electrical parameters
P_{Max} :	Manufacturer-specified maximum power dissipation at T_{A-Max}
P_{App} :	Power dissipation of the device at application temperature limit, T_{App}
P'_{App} :	Power dissipation of the device at the application limit, without margins on the Iso- T_J curve
P_M :	The derating achieved in power dissipation, as a result of the margin put on the Iso- T_J curve. $P_M = P'_{App} - P_{App}$
ΔP :	Change in the power dissipation
V_{CC} :	Supply voltage
I_{CC} :	Quiescent supply current
C_{PD} :	Power dissipation capacitance/buffer
C_L :	Load capacitance/buffer
f :	Frequency of operation of the device
θ_{JA} :	Junction to ambient thermal resistance

B.3 Stress balancing

B.3.1 General

For active semiconductor devices:

$$T_J = T_A + P \times \theta_{JA} \quad (B.1)$$

where

T_J is the junction temperature,

T_A is the ambient temperature,

P is the power dissipation, and

θ_{JA} is the junction to ambient thermal resistance.

If the junction temperature of a semiconductor device remains constant, then the performance of the device should not change. The power dissipation of a device is often a function of some electrical parameter (for example, operating voltage, frequency); thus a trade-off can be made between ambient temperature and an electrical parameter. From Formula (B.1), a higher ambient temperature is allowed if the power dissipation is reduced sufficiently to keep the junction temperature constant. The steps to be followed in stress balancing are listed in B.3.2 through B.3.7, and shown schematically in Figure B.5.

B.3.2 Determine the ambient temperature extremes

The goal of this step is to determine the extreme temperature near the device considering the equipment ambient temperature range and the fact that temperature rise occurs from ambient temperature outside the equipment. This results in a defined ambient temperature environment for the device.

B.3.3 Determine parameter relationship to power dissipation

The goal of this step is to determine which electrical parameters can be derated⁴, and by how much, and to calculate the amount by which the dissipated power should be reduced in the proposed application, in order to uprate the device.

In some circumstances it can be necessary to reduce more than one parameter to obtain the desired reduction in dissipated power.

NOTE Various factors, such as device technology, device family, and electrical function, are considered in selecting the parameters that most significantly affect power dissipation, and therefore are selected for reduction. As an example, the relationship between power dissipation and effective operating frequency is essentially linear for CMOS devices, and changing the device operating voltage can reduce dissipated power. Other possibilities include reducing the output current, reducing the fan-out, or altering the duty cycle of the device.

B.3.4 Determine the dissipated power versus ambient temperature relationship

B.3.4.1 General

The goal of this step is to produce a graphical representation of the relationship between device power dissipation and the ambient temperature, as defined in Formula (B.1). The power dissipation is plotted against the ambient temperature, keeping the junction temperature constant. The Iso- T_J plot, an example of which is shown in Figure B.1, can be constructed using either of the two processes described in B.3.4.2 and B.3.4.3.

NOTE A generalised Iso- T_J curve is shown in Figure B.1. The curve is drawn for power dissipation values that lie between the minimum power dissipation of the device (P_{Min}) and the maximum specified power (P_{Max}). To account for inaccuracies⁵ in the data and calculations, the curve is moved towards the horizontal axis by a suitable amount, T_M . T_M can be viewed as the junction temperature margin. The application power (P_{App}) as calculated from the Iso- T_J curve would also have a reduced value, with a margin P_M as illustrated in Figure B.1. A corresponding temperature range above the maximum operating ambient temperature (T_{A-Max}) is thus obtained. The temperature corresponding to P_{Min} is the maximum temperature at which the device can be used in the application. This is denoted by T_{Up-Max} . As such, the area bounded by $P_{Max} - P_{Min} - I - I'$ is the uprated operating area. Combinations

⁴ Derating is the practice of limiting thermal, electrical or mechanical stresses on electronic parts to levels below the manufacturer's specified ratings. As the term is used here, it may be said that one or more parameters are 'derated' in order to uprate a component.

⁵ The inaccuracies may be in the calculation of power dissipation, in the determination of the thermal characteristics of the part and due to unavailability of accurate thermal resistance of the part.

of power and temperature values in this area correspond to junction temperatures lower than that established by the Iso- T_J curve with margins $I - I'$.

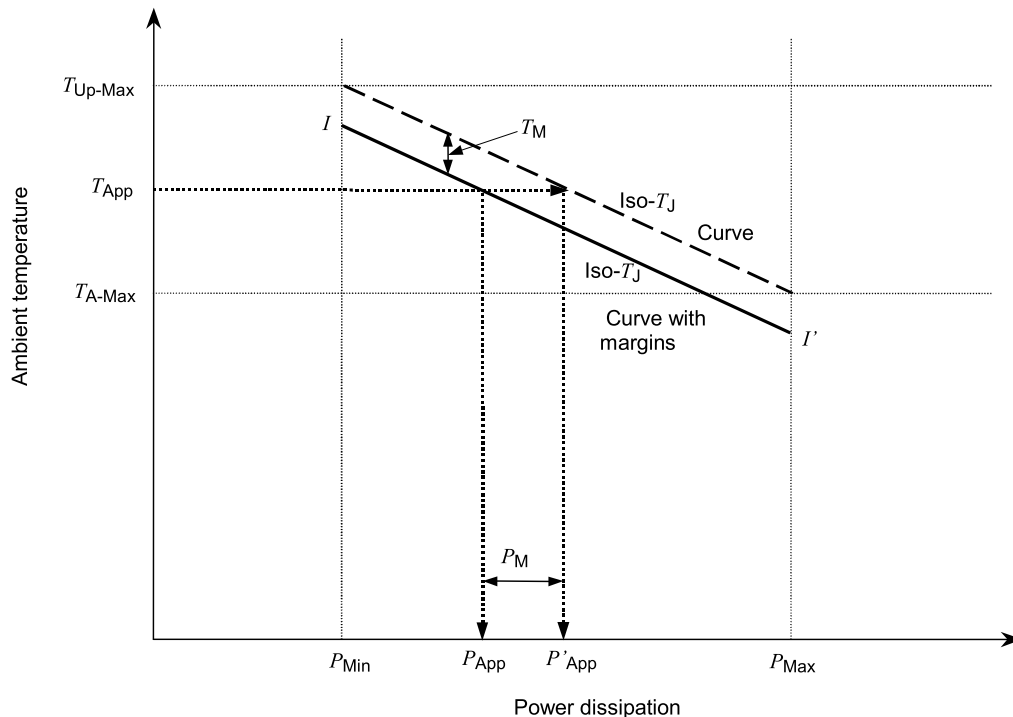
B.3.4.2 Constructing the Iso- T_J curve using thermal resistance

If Formula (B.1) is modified as follows:

$$T_A = -\theta_{JA} \times P + T_J \quad (\text{B.2})$$

then a plot of power dissipation versus ambient temperature yields a straight line with slope, that is, θ_{JA} . If the line passes through the point $(T_{A-\text{Max}}, P_{\text{Max}})$, where P_{Max} is the maximum power dissipation specified by the manufacturer at the specified maximum operating ambient temperature, $T_{A-\text{Max}}$, then it is called the Iso- T_J curve.

To plot the Iso- T_J curve, the junction to ambient thermal resistance should be known. Some data sheets include θ_{JA} in a thermal characteristics section. The thermal resistance value in the application also depends on factors such as thermal conductivity of the printed circuit board, proximity and power dissipation of neighbouring devices, airflow speed and pattern, coolant physical properties, die size, and thermal radiation properties of the surrounding surfaces. Most of these factors impact the thermal impedance from case-to-ambient, and not all of them can be modeled accurately early in the design process. In spite of these difficulties, the electrical and mechanical designers work together to determine the thermal resistance data in the application using the best information available. Whenever possible, the test or simulation conditions under which the thermal resistance data were determined are obtained from the device manufacturer before the thermal resistance data are used. The data used early in the design stage should be verified by testing in the application environment and later in the development process.



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Figure B.1 – Iso- T_J curve: Relationship between ambient temperature and dissipated power

B.3.4.3 Constructing the Iso- T_J curve using thermal analysis

Thermal simulation software may be used to evaluate the performance of the device in the application, provided that its range of applicability includes the required application temperature, power, etc. Typical steps of stress balancing can be:

- a) Develop a thermal model of the device.
- b) Conduct thermal simulation using the device as the “device under test” in the manufacturer’s thermal test setup. The model is valid if the device thermal simulation compares satisfactorily with the thermal data provided by the manufacturer.
- c) Model the application environment with different values of power dissipation within the device.
- d) Develop the power/temperature relationship from the application thermal model.

NOTE The value of thermal resistance also varies with power dissipation and temperature. This is accounted for in the thermal simulation, but not in the simplistic single parameter approach described in B.3.4.2, which assumes θ_{JA} to be constant for the temperature and power dissipation range under consideration. The Iso- T_J curve obtained by thermal analysis or simulation therefore is not necessarily linear, and the thermal characterisation obtained in B.3.4.3 is likely to be more accurate.

B.3.5 Assess applicability of the method

If the required maximum system temperature is lower than T_{Up-Max} , then the device can be uprated by stress balancing. If the required maximum system temperature is greater than T_{Up-Max} , then the required power dissipation is lower than P_{Min} , and other options are to be considered.

A horizontal line drawn through the required ambient temperature (on the vertical axis) intersects the Iso- T_J curve at a power dissipation value (on the horizontal axis) equal to the maximum power (P_{App}) that the device is allowed to dissipate at the application temperature. All selected electrical parameters are then modified to maintain the device power dissipation below P_{App} .

B.3.6 Determine the new parameter values

Figure B.2 shows a generalized plot of an electrical parameter versus the dissipated power for the “allowable” range of power dissipation between P_{Min} and P_{Max} . The vertical line corresponds to the application power dissipation, P_{App} , of the uprated device. The value of the electrical parameter at the point where this line intersects the vertical axis is the value of the electrical parameter, as modified by stress balancing.

NOTE The changes made to the electrical parameters to reduce power dissipation can change other electrical parameters of the device. For example, if the supply voltage of an operational amplifier is reduced, its frequency range and output current will also change and it will saturate at a lower input voltage. These effects are taken into account while designing with the uprated device.

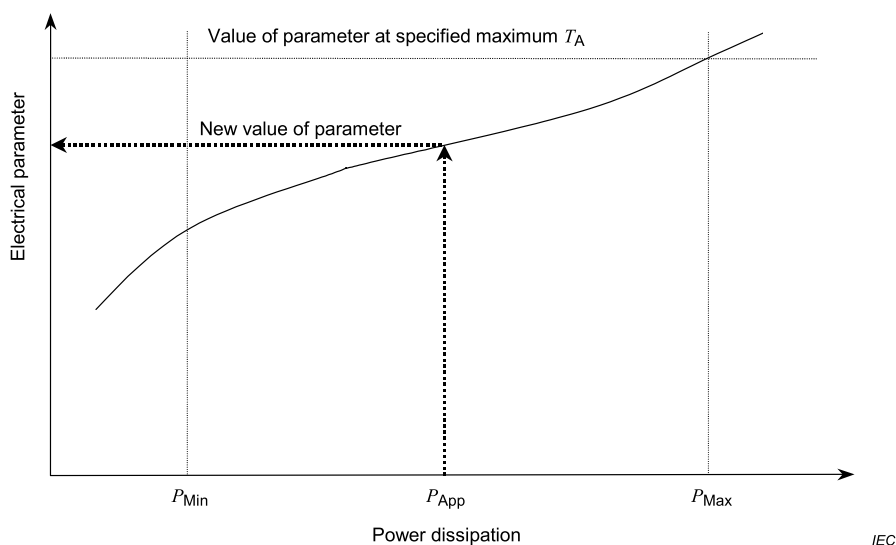


Figure B.2 – Graph of electrical parameters versus dissipated power

B.3.7 Conduct parametric and functional tests

After successful completion of steps B.3.2 through B.3.6, parametric and functional tests are performed at the target application temperature, using the new values of electrical parameters. This is done to:

- ensure that the device and the system operate satisfactorily with the newly calculated conditions;
- verify that the device will operate successfully in the new conditions;
- check for discontinuities and changes in parameter trends in the vicinity of the extreme target application temperature; and
- check the adequacy of margins for the extreme target application temperature and selected derated parameters from B.3.4.

The tests may be done at the device level, or using devices in the application. A device can satisfy its parameter specifications, but still fail to function in an application. Therefore, functional testing in the application or higher levels is considered.

NOTE In choosing the sample size of devices to be tested, consideration is given to margins, confidence testing, and variations in the parameter values. The tests can be parametric go/no-go tests. The device is considered successfully updated for the application if the test results demonstrate that the device can operate successfully in the application over the full target application temperature.

Tests on devices and systems typically are not of sufficient duration to allow thermal equilibrium to be reached. It can therefore be necessary to power up the system to allow the system device to reach thermal equilibrium prior to testing.

B.4 Application example

B.4.1 General

This example is presented courtesy of the University of Maryland, CALCE, from a 1999 paper: *"Stress Balancing: A Method for Use of Electronic Parts Outside the Manufacturer Specified Temperature Range."*

The Fairchild MM74HC244 is used here as an example to illustrate the stress balancing process for a hypothetical system. The maximum application ambient temperature is 85 °C and the minimum application ambient temperature is –40 °C. The MM74HC244 is an octal

three-state buffer, which is typically used to buffer a bus before connecting to input or output devices. It is a CMOS logic device rated for a $-40\text{ }^{\circ}\text{C}$ to $+85\text{ }^{\circ}\text{C}$ ambient temperature range, and is available in a plastic dual-in-line package. The data sheet recommends operation at a supply voltage (V_{CC}) of 2 V, 4,5 V or 6 V. Absolute maximum rating for power dissipation at $65\text{ }^{\circ}\text{C}$ is 600 mW, with a derating factor of $-12\text{ mW}/^{\circ}\text{C}$ above $65\text{ }^{\circ}\text{C}$. This means that the maximum power dissipation of the device at $85\text{ }^{\circ}\text{C}$ (maximum operating ambient temperature limit) is 360 mW. For this example, assume that the digital logic levels of the system are 4,4 V (minimum) for high and 0,1 V (maximum) for low. From the data sheet, this requires $V_{CC} = 6\text{ V}$.

B.4.2 Determine the ambient temperature extremes

The system ambient temperature range is $-40\text{ }^{\circ}\text{C}$ to $+85\text{ }^{\circ}\text{C}$, however it is assumed that a $15\text{ }^{\circ}\text{C}$ temperature rise occurs from ambient temperature outside the equipment to ambient temperature near the device, due to internal heating effects while the equipment is operating. This results in a $-40\text{ }^{\circ}\text{C}$ to $+100\text{ }^{\circ}\text{C}$ ambient temperature environment for the device.

B.4.3 Select the parameters that can be derated

CMOS devices typically have negligible quiescent power consumption compared to the power dissipation during switching. The power dissipation (P) of a CMOS device is given by the formula:

$$P = (C_{PD} + C_L) V_{CC}^2 f + V_{CC} I_{CC} \quad (\text{B.3})$$

where

C_{PD} is the power dissipation capacitance,

C_L is the load capacitance,

f is the switching frequency,

I_{CC} is the quiescent supply current, and

V_{CC} is the supply voltage.

From the Fairchild data sheet:

$C_{pd} = 50\text{ pF/buffer}$ and

I_{CC} (maximum specified, for $V_{CC} = 6\text{ V}$) = $160\text{ }\mu\text{A}$.

The load capacitance (C_L) is assumed to be 50 pF, which is the value used for test conditions in the data sheet.

From Formula (B.3), power dissipation can be reduced either by reducing the supply voltage or the operating frequency. In this application, however, a change in V_{CC} will directly affect the logic levels of the system, so V_{CC} cannot be changed. Thus in this example, the operating frequency is reduced from its maximum capability. The frequency at a power dissipation of 360 mW, at $V_{CC} = 6\text{ V}$ is 13 MHz, and it is assumed that the system requirement is for the device be operated at a frequency no less than 3,5 MHz. This means that the minimum power dissipation is 100 mW.

In CMOS devices, the value of load capacitance, C_L , may not be known precisely. The primary contributor to load capacitance in digital logic devices is the capacitance added due to the printed circuit board (PCB)/printed wiring board (PWB) when the devices are installed.

A conservative maximum estimate of C_L should be made for calculating the power dissipation. After the printed circuit board (PCB)/printed wiring board (PWB) design is done, the actual value of C_L may be re-substituted into Formula (B.3) to obtain a better estimate.

B.4.4 Construct an Iso- T_J plot

An Iso- T_J curve for the MM74HC244 plotted from the Fairchild data sheet is shown in Figure B.3. The specified maximum power is 360 mW at a maximum ambient temperature of 85 °C, so the curve passes through the (360 mW, 85 °C) point.

B.4.5 Determine whether or not the device can be upgraded

The minimum power at which the buffer can operate in the system is calculated as 100 mW (see B.5.2). The slope of the Iso- T_J curve is $-83,3$ °C/W (corresponding to -12 mW/°C power derating), which is higher than the thermal resistance value quoted by Fairchild for this device (61 °C/W). The maximum ambient temperature at which the device can be upgraded (T_{Up-Max}) is 106 °C. At the maximum application ambient temperature (100 °C), it is observed that the new power value is 170 mW. The device can thus be operated at 100 °C, if the power dissipation is kept below 170 mW.

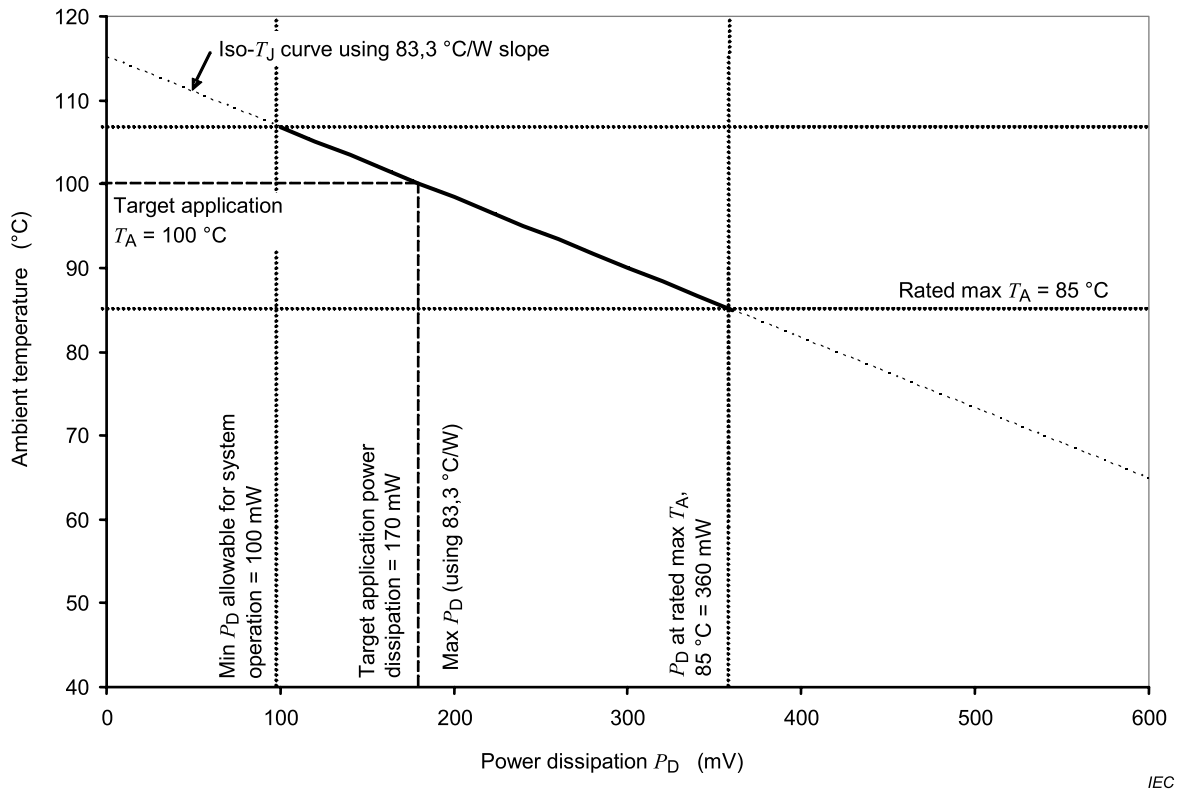


Figure B.3 – Iso- T_J curve for the Fairchild MM74HC244

B.4.6 Determine the new parameter values

As specified in B.4.3, the minimum power at which the buffer can operate in the system is 100 mW and the maximum specified output power is 360 mW. Using Formula (B.3), operating frequency is plotted versus power dissipation, and is shown in Figure B.4. From the data sheet values and application conditions, the formula for the curve is:

$$P = 2,88 \times 10^{-8} f + 9,6 \times 10^{-4} \text{ W} \quad (\text{B.4})$$

Figure B.4 is plotted with the data sheet specified values for the various parameters, with $V_{CC} = 6$ V and $C_L = 50$ pF. It is also noted that there are eight buffers in the MM74HC244N.

From Figure B.4, the frequency is 6 MHz at 170 mW. Thus the device can be used at an ambient temperature of 100 °C, if the frequency is maintained at (or below) 6 MHz. This value may be further derated, depending upon the equipment manufacturer's design practices.

NOTE A relatively simple logic device is used in this example. For more complex devices, the assumption that the maximum operating junction temperature is the average temperature across the die surface may not be accurate, due to hot spots. When possible, this type of information is requested from the device manufacturer, or an increased margin above the junction temperature maximum plotted from Figure B.1 is used.

B.4.7 Conduct parametric and functional tests

Normally, the device is tested at T_{A-Max} to ensure that the device will operate satisfactorily in the required environment.

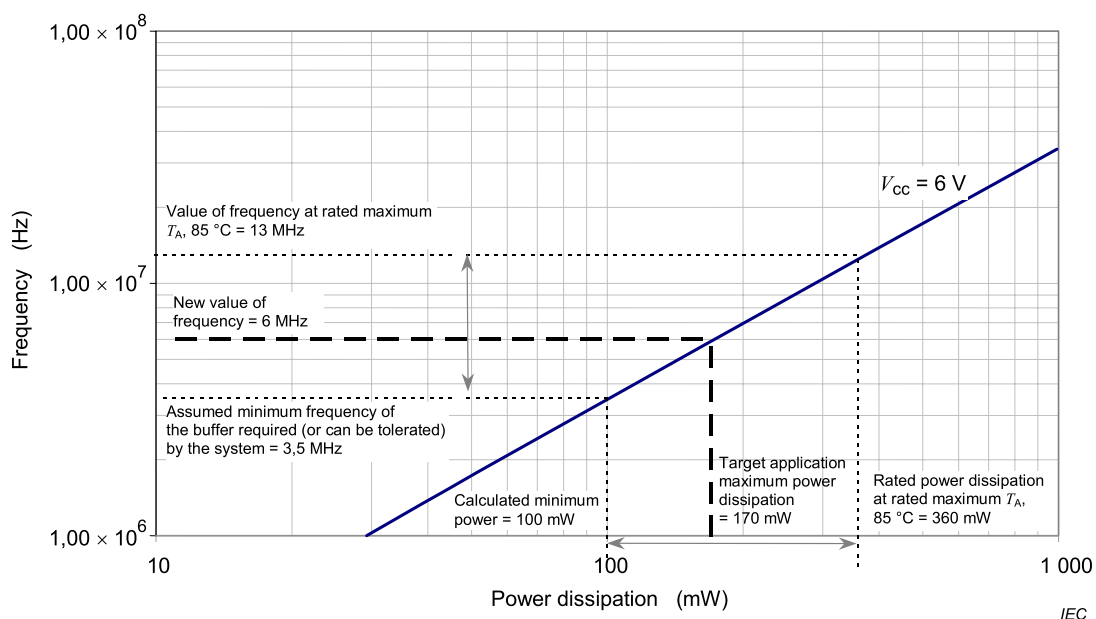


Figure B.4 – Power versus frequency curve for the Fairchild MM74HC244

B.5 Other notes

B.5.1 Margins

The ambient temperature and dissipated power used in the calculations for stress balancing are manufacturer-specified, and have the manufacturer's intended margins. The derated value of power dissipation also has these margins. It can be prudent to add an additional margin (P_M) in calculating the application power P_{App} to compensate for inaccuracies in thermal modeling, value of θ_{JA} and actual ambient temperature near the device in the application. The device electrical parameters, which are calculated from the derated value of power dissipation, are also subject to derating. This is to accommodate the errors inherent in the process; however, adequate precautions need to be taken while using this process.

If a parametric pass/fail test is used to test the devices after stress balancing analysis, the test is carried out at the target application temperature plus an appropriate thermal margin. This will ensure that some margins exist at the target application temperature.

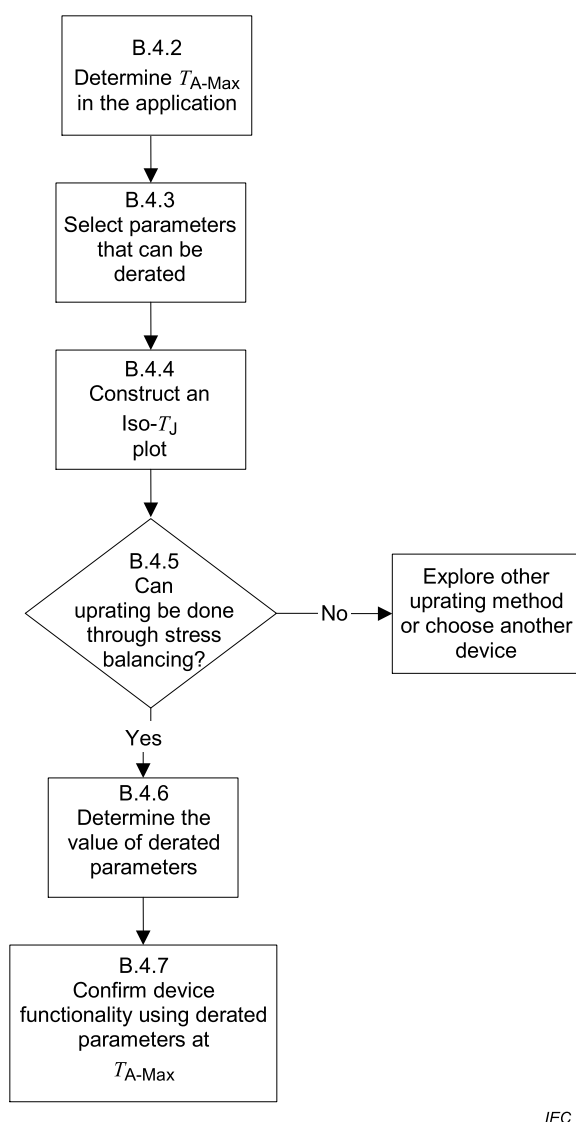
B.5.2 Cautions and limitations

Although stress balancing appears to be straightforward, there are certain hidden difficulties. Data sheet junction temperature limits are used with extreme caution for calculations in stress balancing, because they do not reflect the maximum junction temperature at which the device

would operate. The thermal resistance values of a device are application dependent. If they are used to construct the Iso- T_J curve, thermal validation by test or analysis should also be done.

The power dissipated by the device is the power that is lost as heat, and is not the same as the output power. For CMOS devices, all the power drawn by the device is dissipated, because the output current is negligible. However, for certain devices, data sheets may list only output power. The dissipated power is calculated from additional data obtained from the manufacturer.

Stress balancing can be used for uprating devices above the rated maximum temperature limit only. The relationship given in Formula (B.2) does not apply to evaluating devices for use below the manufacturer-minimum specified temperature limit. Other methods such as parameter re-characterisation may be used for operation of devices below the specified temperature range.



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Figure B.5 – Flow chart for stress balancing

DEVICE STRESS BALANCING REPORT		
Device description: Device manufacturer: Device manufacturer part number: Equipment manufacturer drawing number:		Equipment name: Equipment part number: Program manager: Date:
1.	Rated temperature range (case or ambient):	
2.	Usage temperature range (case or ambient):	
3.	Test specification number(s): Parametric: Functional:	
4.	Parameter(s) chosen for derating:	
5.	Iso- T_J plot (please attach separate sheet):	
6.	Amount of margin (or derating) below junction temperature maximum to be used:	
7.	Power dissipation in device with derated parameter:	
8.	Derated value of parameter, chosen from 4:	
9.	Verification test results agree with calculated value (document results and attach):	
10.	Approvals:	

Figure B.6 – Report form for documenting stress balancing

Annex C (informative)

Parameter conformance assessment

C.1 General

Device parameter conformance assessment consists in evaluating electrical parameters at target thermal test points that are higher or lower than the manufacturer's specified ratings. For this uprating method, the specifications, conditions, and test limits used are the manufacturer's published data sheet parameters.

See the Bibliography for references to statistical methods which are suggested as tools in determining the statistical confidence in the options listed in Annex C.

C.2 Test plan

C.2.1 General

A test plan defines the required tests, process steps, test methods, and number of samples. Figure C.3 shows a flow diagram of parameter conformance assessment testing.

C.2.2 Critical parameters

All critical application parameters are identified, and values obtained from the manufacturer's published data sheet. Ideally, all electrical parameters are tested; however, this is not always possible or practical. For example, it can be difficult to conduct a complete functional test if the applicable test program is not available from the device manufacturer. The manufacturer's specified test parameters (with the exception of the test temperature limits) are used as device performance specifications for parameter conformance assessment.

C.2.3 Minimum allowable test margins

Factors to be considered in developing the test plan for parameter conformance assessment are the specified temperature range of the device, the target temperature range, and previous experience with the device. The test plan includes the temperatures at which tests are conducted, and the test sample size for each temperature. Target temperature information may be obtained from the initial assessment of the equipment's environmental requirements and the results of thermal analysis. If so indicated by the initial assessment and thermal analysis, fluid dynamic conditions, such as air speed and direction, are considered in determining the target temperature and test conditions. All available thermal environment data should be used to calibrate the test equipment to represent the application environment. The type of temperature specification, for example, ambient, case, or junction, is considered in selecting the test temperature range. The accuracy of the thermal assessment method and the test also should be considered.

These test margins provide additional confidence in the applicability of the test results over the target temperature range. Figure C.1 shows the relationship of the various temperatures.

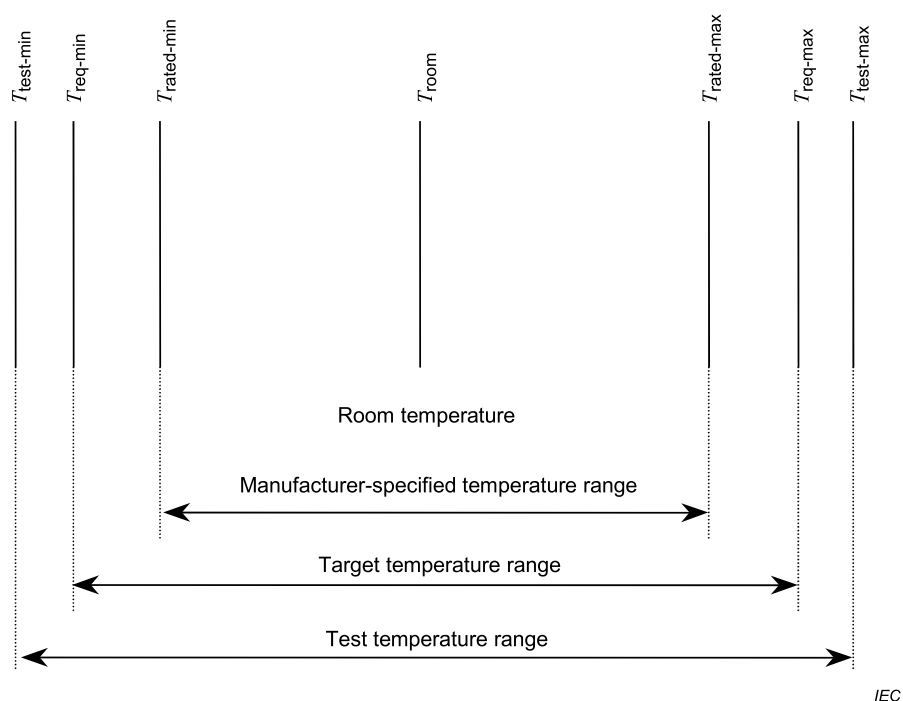


Figure C.1 – Relationship of temperature ratings, requirements and margins

C.2.4 Test options

C.2.4.1 General

The two options for parameter conformance assessment are a): test at the minimum allowable margin (C.2.4.2), and b) determine the margin by incremental temperature testing (C.2.4.3). The selected option is followed by the appropriate quality assurance process to assure that future devices will exhibit the required parameter values at the target temperatures. During either test option, the device's absolute temperature limits should be known and understood. Exceeding any of these absolute limits during either process is controlled and performed only to provide additional understanding of device behaviour and related margins. Devices used in actual applications are not exceeding absolute maximum ratings. A device can satisfy its parameter specifications, but still fail to function in an application. Therefore, functional testing in the application or higher levels is considered.

C.2.4.2 Test at the minimum allowable margin

In this option, parameters are tested at target temperatures above or below the maximum or minimum specified temperature limits. Adequate margins are added to (for target temperature limits above the specified maximum) or subtracted from (for target temperatures below the specified minimum) the target temperature limits.

NOTE Typical margins are 2 °C to 5 °C.

Sample sizes for this option are large enough to provide the desired statistical confidence that the parameter conformance process is successful. The required confidence level, its method of calculation, and results of the calculations are documented for each device uprated by parameter conformance assessment.

Parametric pass-fail tests are conducted for all critical electrical parameters being assessed. The parameter test limits are those of the device manufacturer's data sheet. If this process is successful, then the quality assurance process of C.2.5 is followed. If the process is

unsuccessful, then either the device should not be considered uprateable, or another uprating process should be considered.

C.2.4.3 Determine the margin by incremental temperature testing

C.2.4.3.1 General

In this option, devices are tested at or near the maximum (or minimum) specified temperature limit and then at successively higher (or lower) temperature increments until the parameter of interest no longer conforms to the limits of the data sheet. The recommended temperature increment is 5 °C. The distribution of the temperatures at which non-conformances are observed, and the highest (lowest) temperature limits are determined from these tests.

If

$$T_{\text{req-max}} < (X - CI_X) - A \times CL_\sigma - TE \quad (\text{C.1})$$

where

X is the sample mean,

CI_X is the confidence interval of the mean,

A is the number of standard deviations for the margin,

CL_σ is the confidence level for standard deviation, and

TE is the margin to account for test equipment error,

then the device is capable of being used in the application in question.

C.2.4.3.2 Example: Determine the margin by incremental temperature testing

C.2.4.3.2.1 General

The following conditions are assumed for this example:

- device temperature rating of 0 °C to 70 °C,
- the application ambient temperature range near device is –40 °C to +85 °C,
- the initial test sample is 10 devices,
- the mean fallout temperature of the 10 devices is 130 °C,
- the standard deviation of the mean fallout temperature for the 10 devices is 6 °C,
- the confidence level chosen for this application is 95 %,
- the margin or "A" term in Formula (C.1) is chosen to be 4σ. (Test equipment error is not accounted for in this example.)

Sample sizes for this option are determined on the basis of the observed type of statistical distribution of non-conformance temperatures, the parameters of the distribution, the desired temperature margin, and the desired statistical confidence in the results. All of this information is included in the documentation for each instance of this activity. Figure C.2 illustrates the relationship between the mean fallout temperature, $T_{\text{req-max}}$ and a typical curve representing the device fallout probability at each temperature.

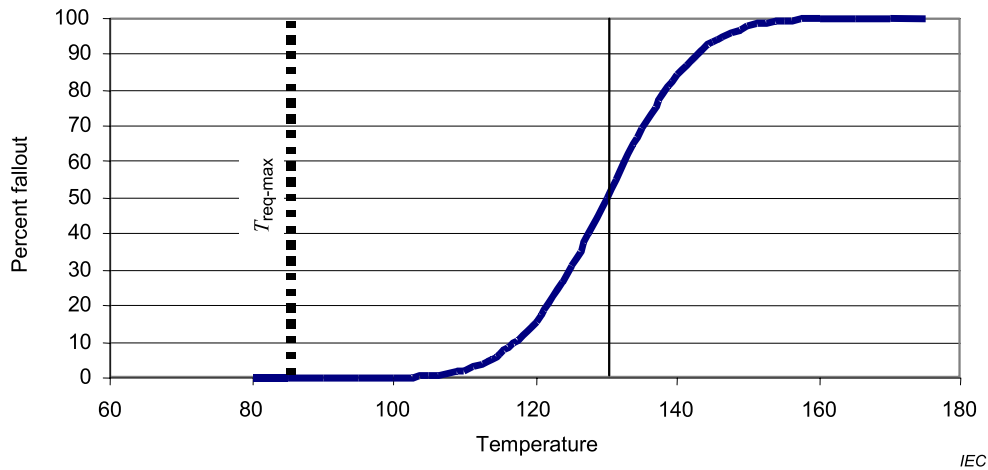


Figure C.2 – Typical fallout distribution versus $T_{\text{req-max}}$

C.2.4.3.2.2 Confidence interval for mean fallout temperature

The formula for calculating the one-sided confidence interval for the mean when the variance is unknown, is

$$CI_X = t_{\gamma, n-1} \times \frac{S}{\sqrt{n}} \quad (\text{C.2})$$

where

$t_{\gamma, n-1}$ is the percentile of the t distribution,

γ is the confidence limit,

$n-1$ are the degrees of freedom

n is the sample size, and

S is the standard deviation.

For this example, the following confidence interval would result:

$$3,478 \text{ °C} = 1,833 \times \frac{6 \text{ °C}}{3,16}$$

C.2.4.3.2.3 Confidence level for standard deviation of mean failure temperature

The formula for calculating the confidence level for the standard deviation is:

$$\sigma_u^2 = (n-1) \times \frac{S^2}{X^2_{1-\gamma, n-1}}, \quad (\text{C.3})$$

where

σ_u^2 is the variance, and

$X^2_{1-\gamma, n-1}$ is the percentile of the X^2 distribution

Formula (C.3) actually computes the upper confidence interval of the variance (σ_u^2).

The confidence limit for the standard deviation is the square root of the variance of the confidence limit. For this example, the following confidence limit would result:

$$9,86 \text{ }^{\circ}\text{C} = \sqrt{9 \times (6 \text{ }^{\circ}\text{C})^2 / 3,33}$$

C.2.4.3.2.4 Result of example

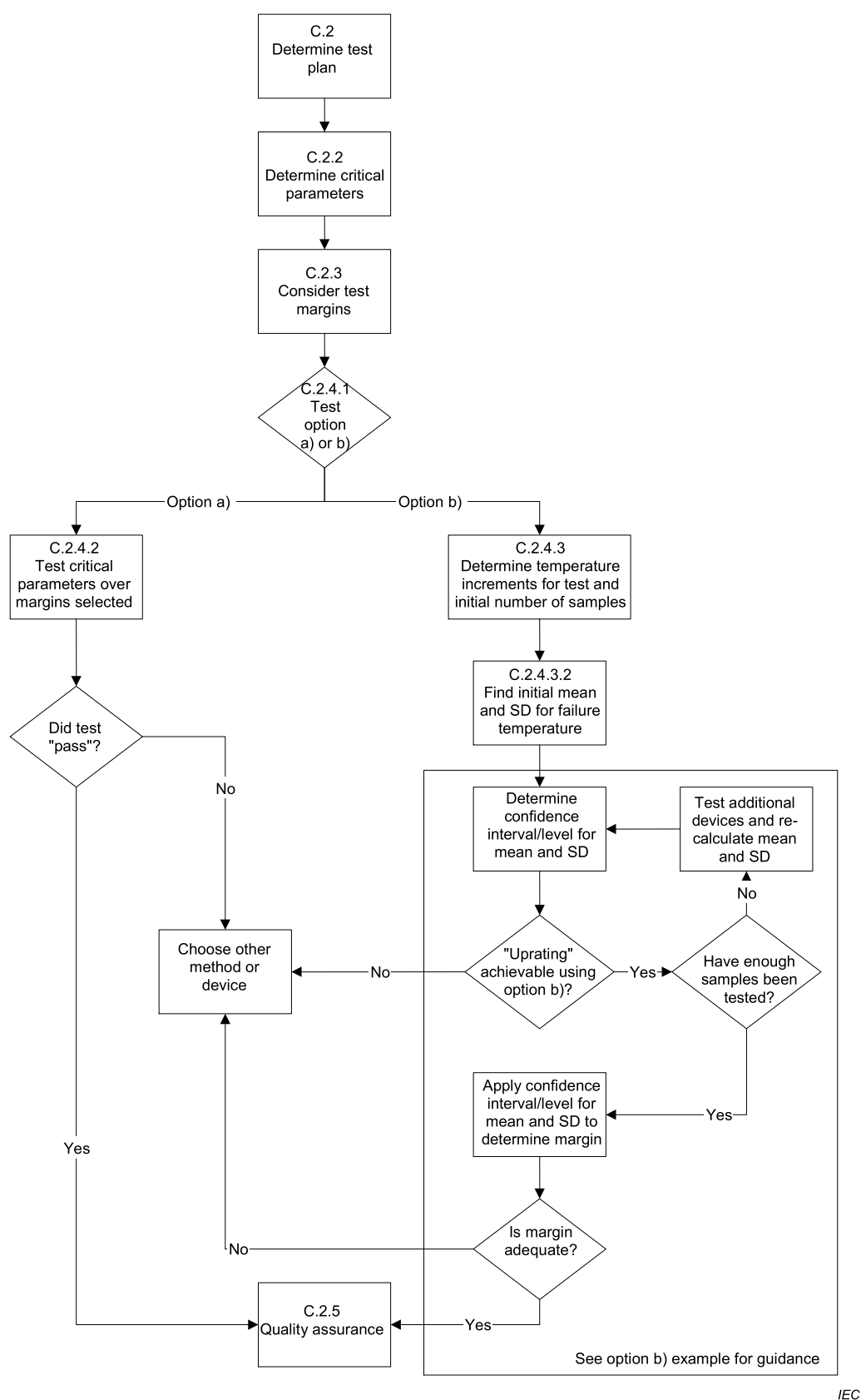
From Formula (C.1):

$$87,07 \text{ }^{\circ}\text{C} = (130 \text{ }^{\circ}\text{C} - 3,478 \text{ }^{\circ}\text{C}) - 4\sigma \times 9,86 \text{ }^{\circ}\text{C}$$

This means that within a proportion of 95 % confidence, not more than 60 parts per million (PPM) of these devices will not perform as per the manufacturer's specification at or below 87 °C. If this capability is sufficient for the application under consideration, the device can be used in the application with some ongoing device or product monitor's reference as in C.2.5.

C.2.5 Quality assurance

Quality assurance may be established by device level testing according to 4.4.2 prior to assembly or higher level assembly testing according to 4.4.3.



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Figure C.3 – Parameter conformance assessment flow

PARAMETER CONFORMANCE TESTING REPORT (to be completed for each lot)		
Device description: Device manufacturer: Device manufacturer part number: Equipment manufacturer part number: Sampling plan reference:		Incoming lot traveller reference: Date:
1.	Test specification number(s): Parametric: Functional:	
2.	Device date code(s):	
3.	#Sample size:	
4.	#Functional and parametric test passed: (Record all test temperatures, and list results for each temperature separately)	
5.	#Functional or parametric test failed: (Record all test temperatures, and list results for each temperature separately)	
6.	Test date:	
7.	Test results report:	
8.	Approvals:	

Figure C.4 – Report form for documenting parameter conformance testing

Annex D (informative)

Higher assembly level testing

D.1 General

Upgrading by assembly test is a method of thermally upgrading devices by testing the assemblies that contain them. Tests are conducted at the assembly level to verify by test correlation that the devices function satisfactorily at the target application temperature range.

This process is to be used to demonstrate the capability of a device to provide the needed function and performance as it is applied in a larger assembly containing other devices and functions, over the target application temperature range. This does not imply assessment of full device capability for all its specified performance characteristics. Only those performance characteristics that are important to the proper performance in the application in which it is used are assessed. Thus, devices that demonstrate acceptance in one application are not automatically approved for other applications.

Typically, assembly level tests are designed to test the basic functional performance of an assembly or device. All functions or key characteristics of the end product are checked at some point in the quality assurance process, whether at the application circuit card level or end product level or somewhere in between. The difference between the typical case and the process described here is that the device's role in these functions, or "key characteristics", of the assembly should be traced and capability verified by assembly test over the target application temperature range.

Figure D.1 illustrates the flow chart of higher level assembly testing.

D.2 Process

D.2.1 General

If assembly level testing is used, the following process should be followed.

D.2.2 Analysis of assembly test definition

The major steps of the assembly test process are:

- a) Determine minimums, ranges and maximums of all functional characteristics of the assembly with respect to the device being upgraded. (Failure modes and effects analysis, fault tree analysis, etc., may be used.)
- b) Assure that the test stimulates and monitors the functional and performance characteristics of the assembly identified in a).

D.2.3 Perform assembly test

Perform the test on 100 % of the assemblies over the target application temperature range, (see IEC TS 62239-1 for test margins). The testing performed at this level should be able to determine whether the device or devices being upgraded adversely affect performance or functionality of the assembly.

NOTE This test can be a separate test or can be included in the acceptance test procedure.

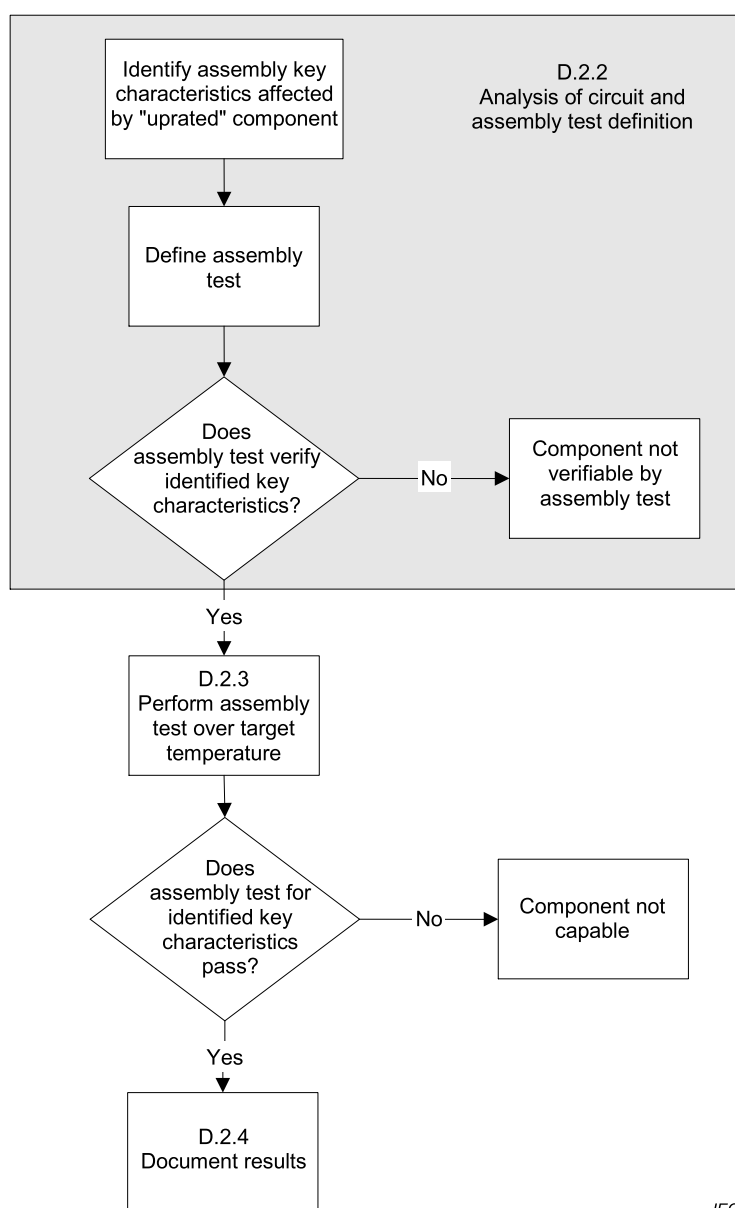
D.2.4 Document results

Document all information from the assessment of the device's function in the assembly, the assembly's key characteristics, the assembly test coverage (related to the device being uprated) and the test results.

D.2.5 Maintenance notification

The steps for maintenance notification are:

- Identify the assembly as one that contains devices uprated by assembly level test.
- Notify all maintainers (and others with a need to know) of the assembly that it contains devices uprated by assembly level test.
- Prepare and provide appropriate information necessary to facilitate the maintenance and other logistics functions.



IEC

Figure D.1 – Flow chart of higher level assembly testing

HIGHER LEVEL ASSEMBLY TEST AT TEMPERATURE EXTREMES REPORT		
Device description: Device manufacturer: Device manufacturer part number: Equipment manufacturer drawing number:		Equipment name: Equipment part number: Program manager: Date:
1.	Rated temperature range (case or ambient):	
2.	Usage temperature range (case or ambient):	
3.	Test specification number(s): Parametric: Functional:	
4.	Determine key characteristics of assembly:	
5.	Determine "uprated" device's role in each key characteristic:	
6.	For each key characteristic upon which the "uprated" device has an effect, document that relationship and effect and ensure that the key characteristic is verified at the application extreme temperature:	
7.	Verify the device using the assembly test at the wider temperature range plus an appropriate temperature margin:	
8.	Did all key characteristics identified in step 6 pass?	
9.	Document results of steps 5, 6, 7 and 8:	
10.	Approvals:	

Figure D.2 – Report form for documenting higher level assembly test at temperature extremes

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